Behaviour Under Addiction: Dynamic Choice Between Smoking and Vaping*

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Abstract

E-cigarettes are addictive but less harmful than combustible cigarettes. Taxes on e-cigarettes may discourage vaping, but could increase more harmful cigarette demand through substitution. I study this policy trade-off using a dynamic discrete–continuous model of joint demand. Households choose which products to buy and how much to purchase. Two product-specific addiction stocks link past purchases to current choices and evolve with the quantities consumed. The model is estimated using NielsenIQ household panel data from 2019 to 2024 by maximum likelihood. A surrogate neural network approximates continuation values across states and parameter values, making estimation feasible. Addiction plays a central role in determining how households respond to price chnages, differently on participation and on quantity choices. Cigarettes and e-cigarettes are substitutes: a permanent 1% increase in the e-cigarette price raises cigarette quantity by 0.23%. A corresponding increase in the cigarette price raises e-cigarette quantity by 0.24%. Price responses weaken with addiction, and substitution into vaping comes mainly from heavily addicted smokers. Removing e-cigarettes from the market would raise cigarette consumption by about 11%: most vapers switch to abstinence but a fair share switch to smoking, while most dual users drop vaping and compensate with more cigarettes. Removing cigarettes from the market, however, does not increase vaping substantially: almost all smokers stop buying cigarettes, and most vapers keep vaping.

Keywords: e-cigarettes, sin taxes, dynamic discrete–continuous choice, addiction, deep equilibrium network, surrogate neural network.
JEL Codes: D12, H21, I18, I12, L66.

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1 Introduction

E-cigarettes (or vapes) have gained considerable popularity over the past decade. This creates a tension for tobacco tax policy. E-cigarettes can help smokers quit and are widely regarded as less harmful than combustible cigarettes (Hajek et al., 2019; McNeill et al., 2015; Allcott and Rafkin, 2022). However, e-cigarettes themselves are not harmless, and are very addictive. The co-presence of good and bad features has produced a sharp policy tension. A tax that discourages vaping may also increase smoking if the two products are substitutes. Whether e-cigarettes should be taxed heavily or treated differently for their potential harm-reduction role, is an open empirical question with significant public-health consequences. The policy trade-off eventually depends on who responds: a smoker who switches products, a dual user who changes quantities, or a vaper who stops buying either product.

This paper studies how cigarette and e-cigarette taxes affect the participation and consumption quantity, and how the responses vary with addiction. I develop and estimate a dynamic discrete–continuous model with two product-specific addiction stocks. Each month, a forward-looking household chooses cigarettes, e-cigarettes, both, or neither. as well as how much of each product to buy. Each product has its own addiction stock, which grows with the quantity bought and depreciates over time. Households can switch freely among actions.

Addiction affects both participation and quantity. On the participation margin, both smoking and vaping addiction stocks enter the utility of every action, so past use of one product can change the appeal of itself and the other. On the quantity margin, each product's own stock shifts the marginal utility of that product and the curvature of its utility. The two margins are linked over time: the quantity bought this month determines how much the stock grows, so a household that buys more today is more addicted tomorrow. A forward-looking household takes this into account when choosing consumption quantity. Quantity margin matters for tax policy since price increases can reduce consumption without changing participation. For example, a heavy smoker could cut down smoking quantity rather than quits. Spending on either product lowers utility through a monetary cost term. There is no direct health cost in the model, but the quadratic terms of addiction stocks capture this cost by allowing the utility of participation to rise with the stock at a decreasing rate and to eventually fall.

I estimate the model using the NielsenIQ Consumer Panel (NielsenIQ, 2024a), which records the cigarette and e-cigarette purchases of a sample of representative U.S. households. The estimation sample contains 6,538 single-person households under the age of 50, observed monthly between 2019 and 2024. Over this period, monthly vaping par-

ticipation more than doubled, from 0.34% to 0.89%, while smoking participation stayed between 5% and 7%. Prices are built at county-month level using NielsenIQ retail scanner data (NielsenIQ, 2024b). Vaping is a fast growing and changing market. The prices are set based on local demand conditions that the econometrician does not observe. To control for this, I use a dynamic control function built from tax-instrumented price innovations (Petrin and Train, 2010). The control function terms enter the utility of vaping, the marginal utility of e-cigarettes, and the state that households use to forecast continuation values. Section 5.4 describes the prices and Section 4.1 the control function.

One challenge on estimation is that the addiction stocks at the start of the panel are not observed. Two households with the same purchases in the panel can have very different pre-panel stocks, and setting the initial stocks to zero for everyone biases the estimated effect of addiction towards zero. I therefore assign each household to one of four latent *types* indexed by its initial-stock profile (Q^s_{i0}, Q^v_{i0}): τ_0 (no prior use), τ_s (prior smoker), τ_v (prior vaper), and τ_{sv} (prior dual user). The type probabilities that depend on demographics and are estimated jointly with the preferences. Types also differ in their action-specific intercepts in utility functions, because permanent preferences are likely correlated with initial addiction. Throughout the paper, I use "prior X" to refer to initial types and verbs ("smoke", "vape", "dual-use", "abstain") for the choice in month t. Section 4.2 describes a worked example, the notation (Table 2), and the construction of initial stocks.

Solving the model is computational difficult. The state has five continuous variables including two prices, two addiction stocks and the e-cigarette control function. Combined with demographic variables, the curse of dimension hits very quickly and ver strongly. This makes conventional value function iteration on a grid less desirable. Deep equilibrium nets (as in (Azinovic et al., 2022)) avoid the grid by training a neural network on simulated states. A network that is trained at one parameter value, however, has to be retrained at every trial value of the updated parameters. This makes the likelihood slow to evaluate, and the potential jumps in value function from re-training mean that the value function can be non-smooth in the parameters. Following the surrogate network approach (Norets, 2012; Kase et al., 2022; Chen et al., 2026), I treat the parameters as additional inputs and train one network on states and parameters together. The major benefit of this method is that once the neural net is trained, solving the model at a new parameter value takes a single forward pass. The likelihood is then smooth and differentiable, and the parameters can be estimated directly without the Expectation–Maximisation algorithm. Section 4.4 describes the method.

The estimates show that addiction works on both the participation choices and quantity choices, and the effects change as the model becoms dynamic. In a static setting, for

cigarettes, the participation utility of smoking has an inverse U-shaped relation with the cigarette stock: it rises from low to median addiction stocks, and falls below its zero-stock level at high stocks. For heavy smokers, the pull towards smoking therefore comes mainly from the quantity margin. The quantity shifter, which scales the marginal utility of cigarettes, is 18 times larger for a heavy smoker than for a new smoker, and the desired quantity is about 12 times larger. For e-cigarettes, the utility of vaping follows a similar inverse U-shape. The quantity shifter of a heavy vaper is 4.5 times that of a new vaper and the desired quantity about 2.4 times.

In a dynamic setting where households take into consideration of the future addiction impact, the numbers change significantly. For heavy users, most of the pull towards using the products comes from the participation margin. A heavy smoker is 40 times more likely to smoke than a non-smoker, and a heavy vaper is 20 times more likely to vape than a non-vaper. Optimal quantity choices reverse here. Both heavy smokers and heavy vapers' optimal consumption quantities are about half of the non-users. This is because in the dynamic setting, consuming too much tobacco in one period comes with significant costs in the future. These include the potential health costs and higher monetary costs. A forward-looking indvidual takes this into account when choosing the consumption quantity, and haevy users intentially reduce the consumption quantities although they are much more likely to keep consuming.

Across products, each stock lowers the utility of using the other product alone. A bulk purchase last month lowers the value of buying more this month, indicating the existence of inventory building behaviours. Curvature parameters on both utility functions exceed one, indicating inelastic responses on the quantity margin, with static price elasticities of about -0.85 for cigarettes and -0.7 for e-cigarettes.

The model reproduces vaping participation and the growth of vaping over time, but it under-predicts smoking participation. The shortfall comes from heavy smokers where predicted smoking differ from the data by the most. The model also over-predicts dual use. The quantity margin fits poor: predicted quanitities barely change with the stock and the model over-predicts the purchases of light smokers and under-predicts those of heavy users of both products.

In counterfactual experiments, cigarettes and e-cigarettes are substitutes, and the responses differ sharply by addiction profile. Holding the stocks at their observed values, a permanent 1% increase in the cigarette price lowers cigarette quantity by about 1.1%, and a 1% increase in the e-cigarette price lowers e-cigarette quantity by about 1.7%. The cross-price elasticities are similar: a higher e-cigarette price raises cigarette quantity by 0.23%, and a higher cigarette price raises e-cigarette quantity by 0.24%. Price responses weaken as the stocks grow, and the substitution into vaping comes mainly from heavily

addicted smokers. Removing e-cigarettes from the market would send most vapers to abstention and a fair share of them to smoking, and most dual users to smoking only, raising cigarette consumption by about 11%, whereas removing cigarettes would make almost all smokers stop buying cigarettes, leave most vapers vaping, and raise vaping participation by only about 3.5%.

The paper makes four contributions. First, it provides a dynamic structural model of joint cigarette and e-cigarette demand, estimated on household panel data, with two interacting product-specific addiction stocks that act on both participation and quantity. Two stocks let a tax on one product affect demand for the other both temporarily and through the gradual evolution of addiction. Second, it shows how addiction operates on both participation and quantity margin, and how the reponses on both margins differ based on the levels of addictions. Third, the results show the importance to consider the addiction profiles when designing tobacco policies, as people's responses can be very different based on their addiction levels. Policies that account for addiction profiles could therefore be targeted at the households whose behaviour a tax actually changes. Fourth, it shows how a surrogate network that takes the parameters as inputs makes maximum likelihood feasible for a dynamic discrete–continuous choice model with latent types.

The rest of the paper is organized as follows. Section 2 reviews the related literature. Section 3 presents the model, Section 4 the estimation method and Section 5 the data. Section 6 reports the structural estimates. Section 7 reports the counterfactual tax experiments. and Section 8 concludes.

2 Related Literature

Multiple addictions. In the rational addiction model of Becker and Murphy (1988), current consumption builds a stock that raises the marginal utility of future consumption. Forward-looking consumers respond to expected future prices (Chaloupka, 1991; Becker et al., 1994). Gruber and Kőszegi (2001) show that present-biased consumers produce the same dynamics, with different welfare implications. However, most of this work has one good and one stock. Bask and Melkersson (2003) extend the model to two goods, cigarettes and Swedish snus, and estimate negative cross-price elasticities on Swedish aggregate data. Bask and Melkersson (2004) apply the two-good model to alcohol and cigarettes, and Andersson et al. (2006) use it to ask whether smokeless tobacco should be promoted in cessation programmes and whether cigarette taxes should rise. Pierani and Tiezzi (2009) estimate a multi-commodity habit model of alcohol and tobacco using Italian data and find that the two goods are complements, although past consumption of one does not reinforce current consumption of the other. These papers show that cross-

stock effects shape how a tax on one good moves demand for the other. Because they use aggregate time series, they cannot tell whether a tax changes how many people use a product or how much each user consumes, nor which users switch. At the household level, Arcidiacono et al. (2007) estimate a dynamic model of discrete drinking and smoking choices, Gordon and Sun (2015) a dynamic model of cigarette addiction with stockpiling on scanner data, and Chen and Rao (2020) a model of cigarettes and e-cigarettes with stockpiling and learning in which both products feed a single nicotine stock. I instead give each product its own stock, let both stocks enter the utility of each action, and treat the four combinations of use as distinct actions. The model can thus capture more complicated switching dynamics across products.

Participation and quantity. Taxes move demand on two margins. Since Lewit and Coate (1982), studies of cigarette demand have attributed a large share of the price response to the decision to smoke rather than to the amount smoked (Chaloupka and Warner, 2000). Two-part and double-hurdle models (Cragg, 1971; Jones, 1989) fit the two margins with separate equations, which leaves the link between them unexplained. Discrete–continuous models derive both margins from one utility function (Dubin and McFadden, 1984; Hanemann, 1984), as do models of multiple discreteness (Hendel, 1999; Kim et al., 2002; Dubé, 2004; Bhat, 2008). Modeling participation and quantity simultaneously comes at a price: the same parameters must explain why some households buy and why some buy a lot. Addiction sharpens this tension. The stock that makes a household more likely to buy also raises the quantity it wants, and the quantity bought today sets tomorrow's stock, so an error on one margin feeds into the other over time. Dynamic models of purchase quantities (Erdem et al., 2003; Hendel and Nevo, 2006) focus on inventory and purchase timing rather than on this link. Gordon and Sun (2015) adapt the addiction into the inventroy-consumption framework to study citarette price elasticity. In my model the stocks shift the two margins through separate channels: action-specific stock terms govern participation, and a stock-dependent shifter governs the marginal utility of quantity.

Neural-network solution and estimation. Full-solution (Rust, 1987) and conditional choice probability (Hotz and Miller, 1993; Aguirregabiria and Mira, 2002; Arcidiacono and Miller, 2011) estimators work on discrete or low-dimensional state spaces. Here the state has five continuous variables, and households also choose continuous quantities, so a giant grid runs into the curse of dimensionality eaily and is not computationally feasible. A growing literature solves high-dimensional dynamic models with neural networks trained on simulated states (Maliar et al., 2021). The deep equilibrium nets of Azinovic et al. (2022) train networks to satisfy the equilibrium conditions along the sim-

ulated paths of the economy. In estimation, a network trained at one parameter value must be retrained at every trial value, which makes the likelihood slow to evaluate and noisy in the parameters. One remedy is to treat the parameters as additional state variables, or pseudo-states (Norets, 2012; Kase et al., 2022; Friedl et al., 2023; Chen et al., 2026). Scheidegger (2026) surveys these methods. I bring this surrogate approach to a household-level discrete–continuous choice model with latent types and a dynamic control function. A second network output, trained on the envelope condition, gives the marginal continuation value that the quantity choice requires.

E-cigarettes and tax policy. Reduced-form studies mostly find that cigarettes and e-cigarettes are substitutes: higher e-cigarette prices or taxes lower vaping and raise smoking (Huang et al., 2014; Zheng et al., 2017; Pesko et al., 2020; Saffer et al., 2020; Pesko and Warman, 2022), although some papers show dual-usage and gateway effects (Owusu et al., 2019; Soneji et al., 2017). These estimates average over users at different stages of addiction and do not separate the immediate response from the adjustment of the stocks. On the normative side, corrective taxes on addictive goods are justified by internalities as well as externalities (O'Donoghue and Rabin, 2006; Gruber and Kőszegi, 2004; Farhi and Gabaix, 2020). Allcott and Rafkin (2022) show that the optimal e-cigarette tax depends on the cross-price elasticity and on whether e-cigarette users would otherwise smoke or abstain. The model here delivers these responses by addiction profile and on both participation and quantity margins.

3 Model

Consumers are assumed to be rational and forward looking. As with the rational addiction literature (Chaloupka, 1991; Becker et al., 1994), the utility of consuming addictive goods depends on past consumpion. Here, addiction stocks for smoking and vaping are measured separately. While nicotine is the main physiologically addictive substance in both products, each product offers unique experience in terms of the flavour, throat hit, sensation and vibe. Having two different addiction stocks in the utility function enables flexible cross-product addiction-dependent consumption, while also allows for addiction to accumulate from nicotine from both products. For example, conditional on the same nicotine addiction level, a chronic smoker can have stronger preference to smoking due to factors other than nicotine. This chronic smoker may still find vaping pleasing due to the nicotine addiction captured in the smoking addiction stock, but such appeal could be much weaker compared to if they were to have the same nicotine-equivalent vaping addiction.

The model derives participation and quantities from a single utility function. Section 3.1 sets out the flow utility, Section 3.2 the addiction stocks and Section 3.3 the dynamic problem. Section 3.4 explains why unobserved local demand shocks call for a control function, which Section 4.1 constructs.

3.1 Flow Utility

In each month t, household i, who lives in county c, chooses one of four mutually exclusive actions: abstain, smoke, vape or use both. Denote the action set $\mathcal{A} \equiv \{0, s, v, sv\}$. There are two products, indexed by $g \in \{s, v\}$: cigarettes (s) and e-cigarettes (v). Let $\mathcal{G}(a)$ be the set of products consumed under action a: $\mathcal{G}(0) = \varnothing$, $\mathcal{G}(s) = \{s\}$, $\mathcal{G}(v) = \{v\}$, and $\mathcal{G}(sv) = \{s, v\}$, as summarised in Table 1.

Table 1: Actions and the products bought.

Action	Short-hand	Products
$a = 0$	0	none
$a = 1$	s	cigarettes
$a = 2$	v	e-cigarettes
$a = 3$	sv	cigarettes and e-cigarettes

In each month t, household i observes its pre-action addiction stocks $\boldsymbol{Q}_{it} = (Q^s_{it}, Q^v_{it})$, whether it bought each product in the previous month, $\boldsymbol{LP}_{it} = (LP^s_{it}, LP^v_{it})$, the county-level prices $\boldsymbol{p}_{ct} = (p^s_{ct}, p^v_{ct})$, and the other demographic variables. The household then selects an action $a \in \mathcal{A}$ and simultaneously chooses a nonnegative quantity x^g_{it} for each $g \in \mathcal{G}(a)$. Quantities are measured in packs per month for cigarettes and millilitres per month for e-cigarettes. The quantities for the product not consumed are set to 0.

Baseline flow utility of household i in month t choosing action a is:

$$u_{it,a}(\boldsymbol{x}) = \underbrace{\boldsymbol{z}_i'\boldsymbol{\gamma}_a + \zeta_{at} + \mu_{a,k(c)} + f_a(\boldsymbol{Q}_{it}) + \sum_{g\in\{s,v\}} \theta^1_{ag} LP^g_{it}}_{\text{participation}} + \underbrace{\sum_{g\in\mathcal{G}(a)} \left[\Phi^g_{it}\, \phi_g(x^g_{it}; \kappa_{git}) + \theta^{2,g}_a(\boldsymbol{LP}_{it})\, x^g_{it} - \alpha\, \tilde{p}^g_{ct}\, x^g_{it}\right]}_{\text{quantity}} + \varepsilon_{iat} \tag{1}$$

where $\boldsymbol{z}_i$ is time-invariant demographics, including income band, ζ_{at} and $\mu_{a,k(c)}$ are month and county-group effects for each action, x^g_{it} is the quantity, and $\tilde{p}^g_{ct}$ is the county-month price divided by the sample standard deviation of monthly spending on the two products

($113). For numerical stability, the county group $k(c)$ is the county itself if the county has at least ten sample households, and a pooled group for its state otherwise. The action-specific function $f_a(\boldsymbol{Q})$ contains linear and quadratic terms of the standardised addiction stocks $\tilde{Q}^g$. The term ε_{iat} is an i.i.d. Type-I extreme value shock to the payoff of action a.

The terms in $\boldsymbol{LP}_{it}$ capture last month's purchase. The indicator LP^g_{it} equals one if the household bought product g in the previous calendar month, and zero otherwise, including when that month is not observed. The level terms θ^1_{ag} let last month's purchase shift the utility of each action beyond its effect through the stocks. The interaction terms $\theta^{2,g}_a(\boldsymbol{LP}_{it})\, x^g_{it}$, which are linear in the indicators, shift the marginal utility of quantity, with coefficients θ^2_{as} on LP^s_{it} and θ^2_{av} on LP^v_{it} (under dual use, each quantity interacts only with its own indicator). This captures bulk purchases and inventory building: a household that bought many cigarettes last month has less need to buy again this month, which a negative θ^2_{as} captures.

The term $\Phi^g_{it}\phi_g(x)$ is the utility function of the quantity consumed, x^g_{it}, scaled by Φ^g_{it}. The scale depends on addiction stocks:

$$\log \Phi^g_{it} = \psi_{0g} + \psi_g Q^g_{it} \tag{2}$$

which allows for own-product addiction to shift the utility of consumption. Cross-product effects work instead through f_a. And the utility from quantity consumed has a concave form:

$$\phi_g(x;\kappa) = \frac{(x+\gamma_g)^{1-\kappa} - \gamma_g^{1-\kappa}}{1-\kappa} \qquad \kappa > 0 \tag{3}$$

where γ_g is a normalisation term to prevent infinite marginal utility at $x^g = 0$. The curvature depends on the own stock,

$$\kappa_{git} = \kappa_{0g} + \kappa_{1g}\,\frac{Q^g_{it}}{\bar{Q}_g} \tag{4}$$

where $\bar{Q}_g$ is the cap on the addiction stock, so κ_{git} runs from κ_{0g} at an empty stock to $\kappa_{0g} + \kappa_{1g}$ at the stock cap $\bar{Q}_g$ (Section 3.2). The static price elasticity of the desired quantity is about $-1/\kappa_{git}$, which lets the intensive margin become more or less price-sensitive as the household becomes addicted. Abstention is the reference action and its utility is normalised to zero. Since $\phi_g(0) = 0$, a product that is not bought yields no quantity utility. Because prices are normalised, $\alpha > 0$ is the utility of spending one standard deviation ($113) more a month, and $\alpha/113$ is the utility of a dollar.

3.2 Addiction Dynamics

The cigarette and e-cigarette stocks evolve separately. For product $g \in \{s, v\}$:

$$Q^g_{i,t+1} = \delta_g Q^g_{it} + \chi_g(x^g_{it}) \tag{5}$$

where $\delta_g \in (0, 1)$ is the product-specific monthly retention rate and $\chi_g(\cdot)$ is an increasing and smoothly capped transformation of quantity, a smooth version of $\min(x/\bar{x}^g, K)$:

$$\chi_g(x) = K - 0.05K \log\left[1 + \exp\left(\frac{K - x/\bar{x}^g}{0.05K}\right)\right] \quad \text{for } x > 0$$

where $\chi_g(0) = 0$. $\bar{x}^g$ is the median purchase in the estimation sample, so that a median purchase adds one unit to the stock. K is the cap that prevents unusually large recorded purchases from creating implausibly large addiction stocks. A month without a purchase adds nothing to the stock. I set $K = 3$. A household at the cap every month approaches $K/(1-\delta_g)$, which is 7.5 for cigarettes and 10 for e-cigarettes. Few households come close to these levels, especially for e-cigarettes[1].

3.3 Dynamic Decision Problem

Let $\boldsymbol{S}_{it} = \left(p^s_{ct},\, p^v_{ct},\, Q^s_{it},\, Q^v_{it},\, LP^s_{it},\, LP^v_{it}\right)$ denote the collection of state variables. The choice-specific value function is:

$$W_a(\boldsymbol{S}_{it}, \boldsymbol{z}_i) = \max_{\boldsymbol{x}_a \geq \boldsymbol{0}} \{\bar{u}_{it,a}(\boldsymbol{x}_a) + \beta \mathbb{E}\left[V(\boldsymbol{S}_{i,t+1}, \boldsymbol{z}_i) \mid \boldsymbol{S}_{it}, a, \boldsymbol{x}_a\right]\} \tag{6}$$

where $\bar{u}_{it,a}$ is Equation (1) without the preference shock ε_{iat}, $\boldsymbol{x}_a$ are quantities consumed under a, and β is the monthly discount factor. The expectation integrates over next month's price shocks. The addiction stocks evolve deterministically according to Equation (5), and the purchase indicators are set by the current action: $LP^g_{i,t+1} = \mathbf{1}\{g \in \mathcal{G}(a_{it})\}$.

The preference shocks ε_{iat} are i.i.d. Type-I extreme value distributed. The integrated value function and conditional choice probabilities are therefore:

$$V(\boldsymbol{S}_{it}, \boldsymbol{z}_i) = \log \sum_{a \in \mathcal{A}} \exp\{W_a(\boldsymbol{S}_{it}, \boldsymbol{z}_i)\} + \gamma_E \tag{7}$$

$$P(a \mid \boldsymbol{S}_{it}, \boldsymbol{z}_i) = \frac{\exp\{W_a(\boldsymbol{S}_{it}, \boldsymbol{z}_i)\}}{\sum_{a' \in \mathcal{A}} \exp\{W_{a'}(\boldsymbol{S}_{it}, \boldsymbol{z}_i)\}}. \tag{8}$$

[1] In estimation the stocks are capped at the 99.8th percentile of their sample distribution, $\bar{Q}_s = 7.4$ and $\bar{Q}_v = 3.5$.

For each product $g \in \mathcal{G}(a)$, an interior optimal quantity satisfies

$$\underbrace{\Phi_{it}^{g}\phi_{g}'(x_{iat}^{g*};\kappa_{git}) + \theta_{a}^{2,g}(\boldsymbol{LP}_{it})}_{\text{marginal utility}} = \underbrace{\alpha\tilde{p}_{ct}^{g} - \beta\Lambda^{g}(\boldsymbol{S}_{it}, a, \boldsymbol{x}_{a}^{*})\chi_{g}'(x_{iat}^{g*})}_{\text{money cost and future utility costs}} \tag{9}$$

where $\Lambda^{g} = \frac{\partial \mathbb{E}[V(S_{t+1})|\cdot]}{\partial Q_{i,t+1}^{g}}$ is the derivative of expected continuation value with respect to next month's stock $Q_{i,t+1}^{g}$. When $\beta = 0$, there is a closed form solution for quantity:

$$\log(x_{iat}^{g*} + \gamma_{g}) = \frac{1}{\kappa_{git}}\left[\log\Phi_{it}^{g} - \log\left(\alpha\tilde{p}_{ct}^{g} - \theta_{a}^{2,g}(\boldsymbol{LP}_{it})\right)\right]. \tag{10}$$

When $\beta > 0$, the continuation-value term enters and the condition needs to be solved numerically.

3.4 Endogeneity Issue

The baseline model omits the local demand shocks that are observed by households and firms but not by the econometrician. In practice, advertising, promotions, product availability, and local attitudes toward smoking or vaping may affect both participation and quantities consumed. Firms can also respond to the same conditions when setting prices. These brings challenges to the estimation. Treating Equations (1)–(8) as if prices were exogenous would create biases. Section 4.1 addresses these concerns explicitly.

4 Estimation

The main estimation is carried out using Maximal Likelihood. I first estimate the price process in the reduced form, which gives me the law of motion on the price innovations. Conditional on the first-stage estimates, I estimate the participation, quantity, and latent-type parameters jointly. The addiction-stock retention rates δ_{g} are calibrated as described below.

4.1 Price Endogeneity and the Dynamic Control Function

It is common to have local demand shocks that are unobservable in the data, such as local advertising, promotions, and shifts in social norms. These shocks are threats to the estimation as sources of endogeneity. Let ξ_{ct}^{g} denote the demand shock for product $g \in \{s, v\}$ in county c and month t. The shocks are persistent and they are observable

to households. Using cigarette consumption as an example (hence dropping the action subscript in utility and product category in shocks), flow utility becomes:

$$u_{it} = \bar{u}_{it} + \xi_{ct} + \varepsilon_{it}, \tag{11}$$

This creates three problems:

Price endogeneity. Firms observe local demand shocks when setting prices, which means $\text{Cov}(\boldsymbol{p}_{ct}, \boldsymbol{\xi}_{ct}) \neq \mathbf{0}$. Consequently, the composite residual $\tilde{\varepsilon}_{it} = \xi_{ct} + \varepsilon_{it}$ is correlated with prices, $\mathbb{E}[\tilde{\varepsilon}_{it} \mid \boldsymbol{p}_{ct}] \neq 0$, and the estimators would be biased.

Distribution of the residual and endogenous stocks. With unobservable demand shock, the residual $\tilde{\varepsilon}_{it}$ is no longer i.i.d. EV1. This invalidates the logit form of the choice probability calculations. In addition, addiction stocks $\boldsymbol{Q}_{it}$ are functions of past choices, which were themselves influenced by past shocks, so persistent shocks induce correlation between $\boldsymbol{Q}_{it}$ and $\tilde{\varepsilon}_{it}$. Omitting them therefore biases not only the price coefficients but also the addiction parameters.

Misspecified expectations. The value function depends on the demand shock $V = V(\boldsymbol{p}_{ct}, \boldsymbol{Q}', \boldsymbol{\xi}_{ct})$. Because the shock is persistent, a positive shock today will change tomorrow's value function, and this will then affect choices today via the forward-looking channel. Similarly, this also creates a positive correlation between continuation value and prices.

Let $\xi_{ct}^{ext,g}$ be an unobserved county-month demand shock that shifts the participation preference (extensive margin), and let $\xi_{ct}^{int,g}$ be the shock to the marginal utility of quantity consumed (intensive margin). The utility can be re-written as:

$$u_{iat}^{\text{true}}(\boldsymbol{x}) = \bar{u}_{iat}(\boldsymbol{x}) + \sum_{g \in \mathcal{G}(a)} \xi_{ct}^{ext,g} + \varepsilon_{iat} \tag{12}$$

and the shifter of the utility from quantity consumption can be written as:

$$\log \Phi_{it}^{g} = \psi_{0g} + \psi_{gs} Q_{it}^{s} + \psi_{gv} Q_{it}^{v} + \xi_{ct}^{int,g} \tag{13}$$

The first shock moves participation directly, as it increases the overall utility from action a, regardless of the quantity consumed. The intensive-margin shock changes the optimal quantity and therefore also the value of participation. If the shocks are persistent, both

will additionally affect the continuation value and its derivative in Equation (9).

Let $\boldsymbol{\tau}_{st}$ be the state taxes on cigarettes and e-cigarettes and $\mathcal{I}_{c,t-1}$ be the information available at the end of month $t-1$.

Assumption 1 (Persistent demand shocks)**.** *For each g, $\xi^g_{ct} = \rho_g \xi^g_{c,t-1} + \eta^g_{ct}$, where $\mathbb{E}[\eta^g_{ct} \mid \boldsymbol{\tau}_{st}, \mathcal{I}_{c,t-1}] = 0$.*

Assumption 2 (Pricing)**.** *The innovation is $\eta^g_{ct} = \eta^{g,f}_{ct} + u^g_{ct}$, where firms observe $\eta^{g,f}_{ct}$ when setting prices but not u^g_{ct}. Prices depend on taxes, policy controls, lagged information, the firm-observed demand innovations, and supply innovations $\boldsymbol{\omega}_{ct}$.*

Assumption 3 (Distribution)**.** *The firm-observed demand innovations $\boldsymbol{\eta}^f_{ct}$ and the supply innovations $\boldsymbol{\omega}_{ct}$ are jointly normal, serially uncorrelated, and independent of $(\boldsymbol{\tau}_{st}, \mathcal{I}_{c,t-1})$.*

One great advantage is that assumption 1 does not require tax levels to be unrelated to persistent demand shocks. It allows taxes to respond to persistent smoking or vaping prevalence, but only required that taxes do not respond to the contemporaneous demand innovation.

At the first stage, I estimate[2]

$$\boldsymbol{p}_{ct} = B_0 \boldsymbol{\tau}_{st} + B_1 \boldsymbol{\tau}_{s,t-1} + \sum_{\ell=1}^{L} B_{2,\ell} \boldsymbol{p}_{c,t-\ell} + \boldsymbol{\mu}_c + \boldsymbol{\zeta}_t + \boldsymbol{v}_{ct} \tag{14}$$

with county and time effects $(\boldsymbol{\mu}_c + \boldsymbol{\zeta}_t)$ absorbing the permanent geographic differences and common price movements. Under the assumptions above, $\boldsymbol{v}_{ct}$ contains the price innovation from the demand shocks.

Re-write the demand innovation η^g_{ct} as a linear function of $\boldsymbol{v}_{ct}$ (as in least squared regression) gives $\eta^g_{ct} = \boldsymbol{\lambda}_g' \boldsymbol{v}_{ct} + \widetilde{u}^g_{ct}$. And iterating the demand-shock backward yields:

$$\begin{aligned} \xi^{ext,g}_{ct} =& \rho_g \xi^{ext,g}_{c,t-1} + \eta^g_{ct} && (15) \\ =& \underbrace{\sum_{\ell \geq 0} \rho_g^\ell \boldsymbol{\lambda}_g' \boldsymbol{v}_{c,t-\ell}}_{C^g_{ct}} + \underbrace{\sum_{\ell \geq 0} \rho_g^\ell \widetilde{u}^g_{c,t-\ell}}_{\mathcal{E}^g_{ct}} && (16) \end{aligned}$$

where

$$C^g_{ct} = \rho_g C^g_{c,t-1} + \boldsymbol{\lambda}_g' \boldsymbol{v}_{ct} \tag{17}$$

[2]The lag length L is chosen to remove residual serial correlation.

The recursively constructed C^g_{ct} is the component of local demand that is correlated with current and past prices. The remainder $\mathcal{E}^g_{ct}$ is orthogonal to the price history under the current assumptions.

To simplify the estimation, I impose the following restriction on the relationship between extensive shocks and intensive shocks:

$$\xi^{int,g}_{ct} = b_g \xi^{ext,g}_{ct} = b_g C^g_{ct} + b_g \mathcal{E}^g_{ct} \tag{18}$$

When $b_g = 0$, the local shocks only affect participation but not optimal quantity. Relaxing equation (18) would require a separate control and transition process for the quantity margin.

In estimation, the fitted control parts $\widehat{C}^g_{ct}$ enter utility function in the place where the demand shocks are:

$$u_{iat}(\boldsymbol{x}) = \bar{u}_{iat}(\boldsymbol{x}) + \sum_{g \in \mathcal{G}(a)} \widehat{C}^g_{ct} \tag{19}$$

$$\log \Phi^g_{it} = \psi_{0g} + \psi_g Q^g_{it} + b_g \widehat{C}^g_{ct} \tag{20}$$

Because the shocks are persistent, the state variable set now expand to include C^g_{ct}:

$$\boldsymbol{S}_{it} = (\boldsymbol{p}_{ct}, \boldsymbol{Q}_{it}, \boldsymbol{LP}_{it}, C^s_{ct}, C^v_{ct}) \,. \tag{21}$$

The transition of the control function term follows equation (17). Its innovations $\boldsymbol{v}$ are drawn jointly with the price innovations $\boldsymbol{r}$ (see Equation (25)) when calculating expectations over future realisations. In the estimated specification only the e-cigarette control function is included ($C^s \equiv 0$), because the estimated cigarette loadings were very close to zero. The control function term's persistence is calibrated as $\rho_v = 0.83$ using auxiliary estimates. This is because it is poorly identified when the loadings are small. After conditioning on $\boldsymbol{C}_{ct}$, the omitted component is only $\mathcal{E}^g_{ct}$, and it is orthogonal to prices. The use of control function, under the current assumptions, removes the source of price endogeneity from the participation margin, the quantity margin, and expected future value.

4.2 Initial Addiction Stocks and Latent Types

The calculation of addiction stocks requires knowing the initial values $\boldsymbol{Q}_{i0}$ when a household enters the panel, but pre-panel consumption is unobserved. Setting both initial stocks to zero would classify long-time users in the same way as new users. This would

attenuate the estimated contributions from addiction to demand. To alleviate this proble, I represent the missing initial condition with four latent types:

$$\mathcal{T} = \{\tau_0, \tau_s, \tau_v, \tau_{sv}\},$$

corresponding to no prior use, prior smoking, prior vaping, and prior dual use, respectively. Note that τ maps households to different initial addiction stock groups, while housholds in each group can still choose any of the four actions $a \in \mathcal{A}$. The distinction is summarised in Table 2.

Table 2: Latent initial-stock types and current actions.

Symbol	Description	Cigarette component	E-cigarette component
Panel A. Latent type τ			
τ_0	no prior use	low	low
τ_s	prior smoker	high	low
τ_v	prior vaper	low	high
τ_{sv}	prior dual user	high	high
Panel B. Current action a			
$a = 0$	abstain	inactive	inactive
$a = s$	smoke	active	inactive
$a = v$	vape	inactive	active
$a = sv$	use both	active	active

To illustrate why initial stocks are important, consider two households that purchase the same cigarette quantity in every observed month, with the addiction accumulation being $\chi_s(x) = 1$. For a given δ_s (e.g. 0.6), household A enters with $Q^s_{A0} = 0$ and household B with $Q^s_{B0} = 1.5$. Their addiction stock paths in the next three months are $(1.00, 1.60, 1.96)$ and $(1.90, 2.14, 2.28)$, respectively. While their in-panel choices are identical, their addiction stocks accumulate very differently. Treating every household with the same initial addiction stock hence introduce considerable measurement errors.

Different ininital addiction stocks could also indicate heterogeneous preferences towards smoking and vaping. To accommondate this, the intercepts inside each action's utility function is allowed to be type-specific:

$$u_{i\tau at} = c_{\tau,a} + \bar{u}_{iat}(\boldsymbol{x}) + \sum_{g \in \mathcal{G}(a)} \widehat{C}^g_{ct}$$

Type probabilities depend on observed household characteristics. Taking τ_0 as the refer-

ence category:

$$\pi_\tau(\boldsymbol{z}_i) = \frac{\exp\{\boldsymbol{z}_i'\boldsymbol{\Gamma}_\tau\}}{1 + \sum_{\tau' \neq \tau_0} \exp\{\boldsymbol{z}_i'\boldsymbol{\Gamma}_{\tau'}\}}, \qquad \tau \in \{\tau_s, \tau_v, \tau_{sv}\} \tag{22}$$

The logit variables $\boldsymbol{z}_i$ contains an intercept, the demographic indicators described in Section 5.5, and supported interactions of income with age and education.

For low initial addiction groups I set $Q^g_{L,0} = 0$ for both smoking and vaping. For high initial addiction groups, the initial stock is the steady-state stock implied by heavy observed use. First, I define the household's average monthly stock increment as

$$\bar{\chi}^g_i = \frac{1}{T_i} \sum_{t \in \mathcal{T}_i} \chi_g(x^{g,\text{obs}}_{it})$$

where $\chi_g(x^{g,\text{obs}}_{it})$ is the stock increment of Equation (5) evaluated at the recorded purchase, so it is zero in months when product g is not purchased. Then

$$Q^g_{H,0} = \min\left\{\bar{Q}_g,\ \frac{\text{P}_{75}(\bar{\chi}^g_i \mid \bar{\chi}^g_i > 0)}{1 - \delta_g}\right\} \tag{23}$$

where $\bar{Q}_g$ is the stock cap of Section 3.2.

4.3 Joint Likelihood

The likelihood has to account for four unobserved objects: the latent type, the local demand shocks, the future price shocks, and the error in recorded quantities. I describe how each one enters, and then combine them into a single objective.

Conditional on type τ, the stock path $\boldsymbol{Q}^\tau_{it}$ follows from the type's initial stocks, the observed purchases, and Equation (5). The type also shifts utility through the intercepts $c_{\tau,a}$. The type is not observed, so the likelihood is a finite mixture over $\mathcal{T}$ with weights $\pi_\tau(\boldsymbol{z}_i; \Gamma)$ from Equation (22).

The control function is built inside the likelihood. Given the first-stage residuals $\widehat{\boldsymbol{v}}_{ct}$ from Equation (14) and the parameters $(\boldsymbol{\lambda}_g, \rho_g)$, Equation (17) gives $C^g_{ct}(\theta)$ for every county-month, starting from zero in the first month of the county series [3]. The loadings $\boldsymbol{\lambda}_v$ and the coefficient b_v are therefore estimated jointly with the preference parameters. The state of a type-τ household is then:

$$\boldsymbol{S}^\tau_{it}(\theta) = (\boldsymbol{p}_{ct},\ \boldsymbol{Q}^\tau_{it},\ \boldsymbol{LP}_{it},\ \boldsymbol{C}_{ct}(\theta)) \tag{24}$$

[3] In practice, I use the price data of the year prior to sample start (2018-2019) to construct the series of CF terms, with the initial period's $C_0 = 0$

Households do not know future prices. They forecast them with

$$p^g_{c,t+1} = \mu_g + a_g\, p^g_{ct} + r^g_{c,t+1} \qquad (\boldsymbol{r}_{c,t+1}, \boldsymbol{v}_{c,t+1}) \sim N(\mathbf{0}, \Omega) \tag{25}$$

The price innovations $\boldsymbol{r}$ and the first-stage innovations $\boldsymbol{v}$ are jointly drawn. This is because a demand innovation potentially moves next month's price and next month's control function at the same time. The expectation in Equation (6) integrates over these four innovations (2 for cigarettes and 2 for e-cigarettes). The parameters (μ_g, a_g, Ω) are estimated in a first step and held fixed.

The recorded purchase quantity is a noisy measure of the quantity chosen in the model. THis is due to factors including discrepancy between purchase and consumption, data aggregation, rounding, and errors in reporting.

$$\log x^{g,\text{obs}}_{it} = \log x^{g*}_{i\tau a_{it} t} + \nu^g_{it}, \qquad \nu^g_{it} \sim N(0, \sigma^2_g) \tag{26}$$

The error ν^g_{it} is independent across households and months, and independent of the utility shocks. It is realised after the household has made its choice.

Let $n(\cdot)$ be the standard normal density. Conditional on type τ, household i's likelihood over all observed months is:

$$L_{i\tau}(\theta) = \prod_{t \in \mathcal{T}_i} P_\tau(a_{it} \mid \boldsymbol{S}^\tau_{it}(\theta), \boldsymbol{z}_i; \theta) \prod_{g \in \mathcal{G}(a_{it})} \frac{1}{\sigma_g\, x^{g,\text{obs}}_{it}}\, n\!\left(\frac{\log x^{g,\text{obs}}_{it} - \log x^{g*}_{i\tau a_{it} t}(\theta)}{\sigma_g} \right). \tag{27}$$

The first term is the action probability in Equation (8). The second term is the density of the observed quantity around the optimal quantity from Equation (9). The same parameters must therefore explain both which products are chosen and how much is purchased.

Using sampling weights w_i, the sample log-likelihood is

$$\mathcal{L}(\theta, \Gamma) = \sum_i w_i \log \left[\sum_{\tau \in \mathcal{T}} \pi_\tau(\boldsymbol{z}_i; \Gamma)\, L_{i\tau}(\theta) \right] \tag{28}$$

where θ collects the utility parameters, the type intercepts, the quantity curvatures, the control-function parameters, and the measurement-error variances.[4] The inner sum is over types: exactly one type applies to each household, so the likelihood of its purchase history is the type-probability-weighted sum of the likelihoods conditional on each type. The outer sum adds the logarithms of these household likelihoods, which is the logarithm of their product across households.

[4] A small ridge penalty, $\frac{1}{2} 10^{-4} \sum_i w_i \|\Gamma_{\text{slopes}}\|^2$, is subtracted from $\mathcal{L}$ to keep the type-logit slopes finite. The intercepts are not penalised.

4.4 Solution and Estimation Algorithm

Evaluating Equation (28) requires the value function at every observed state and every trial value of θ. I first describe the conventional algorithm, then I introduce the surrogate net method.

EM with a deep equilibrium network. The Expectation–Maximisation (EM) algorithm is commonly used to estimate models with latent types (Dempster et al., 1977; Arcidiacono and Jones, 2003). It separates the calculation of posterior type probabilities from parameter estimation, which can simplify maximisation of the mixture likelihood. The E-step computes the posterior probability that household i is of type τ,

$$r_{i\tau} = \frac{\pi_\tau(\boldsymbol{z}_i;\Gamma)L_{i\tau}(\theta)}{\sum_{\tau'\in\mathcal{T}}\pi_{\tau'}(\boldsymbol{z}_i;\Gamma)L_{i\tau'}(\theta)} \tag{29}$$

The M-step updates θ by maximising $\sum_i w_i \sum_\tau r_{i\tau}\log L_{i\tau}(\theta)$, and updates Γ with a multinomial logit of $r_{i\tau}$ on $\boldsymbol{z}_i$. The two steps are repeated until convergence.

Each evaluation of $L_{i\tau}(\theta)$ needs the solution of the dynamic problem. A nested fixed-point estimator (Rust, 1987) would solve Equation (7) on a grid. Here the state has five continuous variables—two prices, two addiction stocks and the control function—besides the two purchase indicators. A grid fine enough to capture the sharp changes in participation and quantity is not feasible. A deep equilibrium network (DEQN) avoids the grid (Azinovic et al., 2022). For a given θ, a neural network $V_\varphi(\boldsymbol{S})$ with network weights φ is trained on sampled states to minimise the gap between V_φ and the Bellman target

$$\mathcal{T}_\varphi(\boldsymbol{S}) = \log\sum_{a\in\mathcal{A}}\exp\left\{\max_{\boldsymbol{x}_a\geq\mathbf{0}}\left[\bar{u}_a(\boldsymbol{S},\boldsymbol{x}_a)+\beta\,\mathbb{E}[V_\varphi(\boldsymbol{S}')\mid\boldsymbol{S},a,\boldsymbol{x}_a]\right]\right\} \tag{30}$$

The network is precise where the data is, because training states are drawn from the observed and simulated state distribution and are not restricted to grid points.

The network above is trained at one value of θ. For each iteration of new θ values, the net needs to be re-trained completely. Traning itself is very computationally costly. This creates two problems.

First, the likelihood is not smooth in θ. Each M-step moves θ, so the network has to be retrained. Training is stochastic, and two networks trained at nearby parameter values differ by training noise. The likelihood then jumps between evaluations, not because θ themselves changed, but because the net has been refit. Numerical Hessians of such an non-smooth objective are unreliable, and so are the standard errors based on them.

Second, the quantity condition (9) depends on the derivative Λ^g of the continuation value,

but the network is trained to fit the level of the value function. A good fit on the level here does not guarantee good fit in the slope. And the noise from a non-smooth net passes directly into the predicted quantities. An alternative method to address the slope issue is the Endogeneous-grid methods as in Carroll (2006); Fella (2014); Iskhakov et al. (2017); Druedahl and Jørgensen (2017). This avoids taking derivatives of an interpolated value function between grid points. However, in my model, the addiction stocks enter the marginal utility, so the Euler inversion does not have a closed form expression. The state space constraint and the need to re-solve the FOC at each trial parameter value makes this method less desirable.

Surrogate network. I therefore treat the parameters as additional state variables and train a single network on states and parameters together. The idea of approximating the solution as a function of both goes back to Norets (2012). Kase et al. (2022) call the parameters pseudo-state variables and Chen et al. (2026) call the network a deep surrogate.

Let $\boldsymbol{S}_{it}^{+} = (\boldsymbol{p}_{ct}, \boldsymbol{Q}_{i,t+1}, \boldsymbol{LP}_{i,t+1}, \boldsymbol{C}_{ct})$ be the state after the quantity choice and before the price innovations are drawn. The network approximates the expected continuation value at this point for any parameter value:

$$\bar{V}_{\varphi}(\boldsymbol{S}^{+}, h; \theta) \approx \mathbb{E}\Big[V(\boldsymbol{S}'; \theta) \mid \boldsymbol{S}^{+}\Big], \tag{31}$$

All the parameters entering the value function enter the net, including: (1) the input θ contains every parameter that enters the Bellman equation: the stock, quantity and curvature parameters, the last-purchase terms, α, and the control-function parameters $(\boldsymbol{\lambda}_v, b_v)$. (2) The type intercepts, the demographic shifters, month effects and the county-group effects enter through the index $h_{i\tau a} = c_{\tau,a} + \boldsymbol{z}_i'\boldsymbol{\gamma}_a + \mu_{a,k(c)}$. One network therefore serves all types and all households.

Training draws states and parameters jointly. States are drawn from a pool of all possible states. Parameters are drawn from a box Θ_B around a starting value.[5] At each draw, the target is the average of the integrated value over N_d joint draws of the innovations,

$$\frac{1}{N_d}\sum_{n=1}^{N_d} \log \sum_{a\in\mathcal{A}} \exp\left\{\max_{\boldsymbol{x}_a \geq \boldsymbol{0}} \left[\bar{u}_a(\boldsymbol{S}_n', \boldsymbol{x}_a; \theta) + \beta\, \bar{V}_{\bar{\varphi}}\Big(\boldsymbol{S}_{n,a}^{+\prime}, h; \theta\Big)\right]\right\}, \tag{32}$$

where $\boldsymbol{S}_n'$ is the n-th draw of next month's state and $\boldsymbol{S}_{n,a}^{+\prime}$ is the post-choice state that follows from it. A second output of the network approximates Λ^g. It is trained on the

[5]The starting value is the estimate at $\beta = 0$, where quantities have the closed form in Equation (10) and no network is needed. The half-widths of the box are multiples of the standard errors at that estimate. If the final estimate lies on the edge of the box, I widen the box and retrain.

envelope condition (i.e. on the derivative of the same target with respect to the stocks). Because the target is an unbiased estimate of the expected continuation value and the network is fitted by least squares over many sampled states, a small fixed number of draws per state is enough: the simulation noise averages out across the training sample, as with the all-in-one expectation operator of Maliar et al. (2021). I use $N_d = 4$. After training, the network weights are fixed. Solving the model at a new θ is a single forward pass.

With the surrogate, $\mathcal{L}(\theta, \Gamma)$ is a deterministic and differentiable function of all parameters. I obtain its exact gradient by differentiating and maximising equation (28) directly over (θ, Γ) with a quasi-Newton method. The type-logit parameters Γ are estimated in the same step as θ, so no E-step and no M-step are needed. Every parameter is estimated in one step.

The surrogate removes stochastic retraining from likelihood evaluation and allows the approximate likelihood to be differentiated through the fixed network. It also avoids solving the dynamic problem separately at every trial parameter value. The continuation value and its derivative both respond to θ through the network inputs. Approximation error remains and depends on training accuracy over the states and parameters used in estimation.

4.5 Identification and Calibration

The model is identified brom both participation and quantity choices. Conditional quantity responses identify the curvature parameters $(\kappa_{0g}, \kappa_{1g})$ and the money coefficient α. Differences in purchases after months with and without a purchase, holding the stocks fixed, identify the last-purchase terms. Participation and quantity jointly identify how each addiction stock shifts action utility and marginal utility. Transitions in both margins identify the dynamic importance of stock accumulation. Dual-use observations and switches between smoking and vaping identify the cross-stock effects and the incremental utility of consuming both products.

After conditioning on the control functions, price sensitivity is identified from price movements generated by the tax instruments and from price innovations that are orthogonal to the projected demand shocks. The exclusion restriction is that, conditional on the policy controls, the county and time effects, taxes affect household utility only through prices and not through contemporaneous demand innovations.

The control function parameters are identified from the first-stage residuals. The loadings $\boldsymbol{\lambda}_v$ are identified by how vaping participation moves with the current residual $\widehat{\boldsymbol{v}}_{ct}$, and

b_v by whether the control function shifts conditional e-cigarette quantities in addition to participation. The persistence ρ_v would be identified by how fast the effect of a past residual fades.

Demographic variation identifies $\boldsymbol{\gamma}_a$ and the type-logit coefficients Γ. The early months of each household are the most informative about the latent initial stocks, because the gap between a high and a low initial stock decays at rate δ_g, which becomes indistinguishable after 6 months. Longer sequences separate this temporary gap from the permanent type intercepts $c_{\tau,a}$.

The discount factor is fixed at $\beta = 0.95$. The monthly retention rates are calibrated to $\delta_s = 0.6$ for cigarettes and $\delta_v = 0.7$ for e-cigarettes. Auxiliary regressions of purchase participation on its own lags guide this calibration.[6] The fitted rates range from 0.46 to 0.69 for cigarettes and from 0.62 to 0.76 for e-cigarettes across specifications. The selected values imply habit half-lives of about 1.4 and 1.9 months.

5 Data

5.1 Data Sources

I estimate the model using the NielsenIQ Consumer Panel (NielsenIQ, 2024a), which records the UPC-level purchases and the demographics of a representative sample of about 60,000 U.S. households per year. Prices are calculated using the NielsenIQ Retail Measurement Services (RMS) scanner data (NielsenIQ, 2024b), which record weekly unit sales and prices by store and UPC for participating retailers. State cigarette excise taxes come from Orzechowski and Walker (2024) and standardised state e-cigarette tax rates from Cotti et al. (2023, 2026). The taxes are used as the instruments in the first stage of the control function (Section 4.1).

5.2 Sample Construction

Because purchases are recorded at the household level rather than for individual members, it is unclear who within a household consumes cigarettes and e-cigarettes. To address this, I keep only single-person households, so that purchases can be attributed to one

[6]For each product, I regress the purchase indicator on six consecutive monthly lags, current own price, demographics, and time effects, dropping the first ten observed months. I estimate pooled and household-demeaned versions. Lagged participation acts only through a geometric stock with a linear effect, $b_k = \lambda\delta_g^{k-1}$. I fit this decay over lags 1–4 in logs and over all six lags allowing an additive constant. This diagnostic draws on the auxiliary-moment approach of Hai and Heckman (2026).

person. Furthermore, households whose head is aged 50 or older are dropped from the sample, because vaping is very rare among older adults. States with 30 or fewer sample households are dropped from the sample entirely.

The final estimation sample contains 6,538 households and 183,147 household-months in 1,214 counties of 38 states (see Appendix Table 12 for detailed decomposition). The median household is observed for 23 months. The sample starts from 2019 because vaping was rare in the panel before then. All the descriptive statistics reported below are calculated using the NielsenIQ projection weights.

5.3 Monthly Observations

Because the data record purchases rather than consumption, it must be aggregated over time so that purchases are informative about consumption. In particular, purchases of cigarettes and e-cigarettes are infrequent. A household that buys two cartons at the start of every month and smokes daily would show three zero-purchase weeks afterwards at weekly frequency. These weeks are inventory consuming, not abstention.[7] On the other hand, aggregating to quarterly level could further alleviate this problem but discards most of the price variation. It could also blur the timing of quitting and switching that the addiction stocks are meant to capture. In this paper, household consumption is aggregated to monthly level to balance these two factors.

The monthly action $a_{it} \in \{0, s, v, sv\}$ is calculated as:

- $a = 0$: *abstain* (no cigarette or e-cigarette purchase).
- $a = s$: *smoke* (cigarette purchase only).
- $a = v$: *vape* (e-cigarette purchase only).
- $a = sv$: *both* (positive purchases in both categories).

A household that buys cigarettes early in the month and e-cigarettes later in the same month is classified as $a = sv$. The quantities are the packs of cigarettes and the millilitres of e-liquid bought in a month. Most re-purchases happen within a month. The median re-purchase gap (in weeks) is only one week. Overall, 87% of cigarettes and 83% of e-cigarettes re-purchases happen within a month (Appendix Figure 19).

[7] An alternative is to model purchase timing explicitly. Hendel and Nevo (2006) models inventory and stockpiling of storable goods. The latter mechanism is relevant here, but the thin sample of vaping households limits the information available to identify an additional latent inventory state.

5.4 Prices

The price of each product is built in two steps. First, for each product I regress the log unit prices on UPC and county-month fixed effects[8]:

$$\log p_{jct} = \alpha_j + \gamma_{ct} + e_{jct}. \tag{33}$$

The UPC effect α_j is the permanent relative price of product j, and the average county-month effect γ_{ct} is the price level of county c in month t, holding the product fixed. The county-month price index is the cost of a fixed basket at the fitted prices:

$$P_{ct} = e^{\gamma_{ct}} \sum_j \bar{w}_j e^{\alpha_j},$$

where $\bar{w}_j$ is the national unit share of product j in the base year (2019 for the first window of the chained index described below). The design works great for cigarettes, but for e-cigarettes this creates a problem because many new e-cigarette products enter the market during the sample period. To address this issue, the price indices are chained over two consecutive years, so that the composition of products remains constant within each two-year window. This limits changes driven by product composition (Aizcorbe et al., 2003; de Haan and Krsinich, 2014). Appendix B gives the details on the construction of chained price index.

Second, every household faces the price index of its county. The prices a household itself pays are not used, because they reflect its choice of brands, pack sizes and stores, which is correlated with its tastes, and because they are not observed in months without a purchase. All prices are deflated to 2019 dollars.

Real prices rose over the sample. The average cigarette price increased by 11%, from \$6.90 per pack in 2019 to \$7.65 in 2024, and the average e-cigarette price by 3%, from \$5.14 to \$5.28 per millilitre, after peaking at \$5.61 in 2023. Taxes also changed (Appendix Figure 16). The average state cigarette excise faced by sample households fell from \$1.90 to \$1.75 per pack in real terms (from 26% to 21% of the retail price), because the tax is set in nominal terms. The average e-cigarette tax doubled, from 5.6% to 11.7% of the retail price, as the number of states taxing e-cigarettes rose from 9 states and the District of Columbia in January 2019 to 30 states and D.C. by the end of 2022.

[8] weighting by units sold

5.5 Demographics

Demographic variables enter the flow utility as additive shifters. They are time-invariant and observed at the household level. The vector $\boldsymbol{z}_i$ contains indicators for gender, marital status, age group, education, occupation, race, and income band. Table 3 shows the composition of the sample.

5.6 Smoking and Vaping Patterns

Participation. At household-month level, smoking participation is 5.8% while vaping participation is 0.63%. Dual usage participation is only 0.13% (Table 4). Smoking participation rose from 5.0% in 2019 to 6.6% in 2022 and fell back to 5.5% in 2024. Vaping participation rose steadily from 0.34% to 0.89%. Smoking increases with age but falls steeply with income, from 11.2% for the below $25,000 group to 2.4% for the above $100,000 group. There is little heterogeneity in vaping participation for different income groups. These purchase rates are well below survey prevalence. For example, in the National Health Interview Survey (NHIS), 14.0% of U.S. adults smoked cigarettes in 2019, and 4.5% used e-cigarettes (Cornelius et al., 2020). Part of the reason is under-recording. Households do not record every purchase, and the outlets where much tobacco (gas stations and corner stores) is bought are under-covered. Appendix A compares the two sources, and Appendix C describes a model of under-recording but unfortunately the very thin vaping sample cannot identify it.

Quantities. A smoker purchases on average 9.5 packs (median 4) a month, and a vaper 30.4 mL (median 16). Both distributions are right-skewed (Appendix Figure 17). Quantities rise with age: from households under 30 to those aged 40–49, packs per smoker rise from 7.9 to 10.5 and millilitres per vaper from 19.5 to 37.5. They also rise over time: the mean number of packs per smoker increases from 7.6 in 2019 to 10.6 in 2024, and the median millilitres per vaper from 7.6 to 24.

For cigarettes, quantities do not rise within households. The increase comes from changes in which households buy.[9] Appendix Table 13 reports the full decomposition. For e-cigarettes, both channels matter. Within households, e-cigarette quantity jumps in 2020, dips in 2022–2023 and rises again in 2024. The pooled increase is larger, so changes

[9] I regress the log monthly purchase on year dummies over buyer-months, $\log x_{it} = \sum_{y=2020}^{2024} \beta_y \mathbf{1}\{y(t) = y\} + \mu_i + u_{it}$, clustering standard errors by household. β_y is the change in the average buyer-month relative to 2019. With household effects, it is the change within a household. The difference between the two is the composition effect. For cigarettes, the 2024 estimate is 0.27. without household effects and -0.15 with them.

Table 3: Demographic composition of the estimation sample (%).

	Share (%)
Age group	
Under 25	4.9
25–29	15.3
30–34	32.4
35–39	13.0
40–44	16.4
45–49	18.1
Annual household income	
Under $25K	30.0
$25–45K	22.7
$45–70K	20.6
$70–100K	11.9
$100K+	14.7
Male	42.3
Married	5.2
Employed	80.2
Education	
No college degree	45.9
College degree	34.1
Postgraduate degree	20.0
Race	
White	66.3
Black	15.8
Asian	7.9
Other	10.0
Occupation	
Professional	61.5
Skilled	17.4
Farmer	0.3
Retired / unable to work	3.7
Not working	17.2

Notes: One row per household (N = 6,538), demographics at the household's first sample month, weighted by NielsenIQ projection weights.

Table 4: Participation and monthly purchase quantities.

		Participation (%)			Packs per smoker-month			mL per vaper-month		
	HH-months	Smoke	Vape	Both	Mean	Median	*N*	Mean	Median	*N*
All	183,147	5.80	0.63	0.13	9.5	4.0	10,144	30.4	16.0	1,218
A. By year										
2019	30,145	4.98	0.34	0.04	7.6	3.0	1,496	24.4	7.6	124
2020	29,922	5.83	0.48	0.10	7.9	4.0	1,627	28.4	9.0	140
2021	30,551	5.93	0.57	0.15	9.2	4.0	1,687	27.9	14.0	170
2022	31,119	6.63	0.66	0.14	9.7	4.0	1,830	24.7	14.4	207
2023	31,917	5.77	0.76	0.16	11.4	5.0	1,880	34.7	21.5	271
2024	29,493	5.54	0.89	0.17	10.6	4.0	1,624	35.1	24.0	306
B. By age group										
Under 30	13,044	3.36	0.64	0.09	7.9	3.0	405	19.5	14.4	67
30–39	63,858	5.21	0.48	0.15	8.7	4.0	3,159	24.3	13.0	374
40–49	106,245	7.28	0.77	0.13	10.5	5.0	6,580	37.5	22.0	777
C. By annual household income										
Under $25K	37,157	11.24	0.80	0.23	9.8	4.0	4,289	27.6	18.0	325
$25–45K	43,445	5.59	0.61	0.12	8.5	4.0	2,554	15.1	9.5	280
$45–70K	48,166	3.98	0.57	0.14	10.8	4.0	1,744	36.1	21.6	251
$70–100K	31,738	3.57	0.72	0.09	7.5	4.3	957	16.8	14.4	275
$100K+	22,641	2.37	0.41	0.03	10.4	4.0	600	70.2	72.0	87

Notes: Participation is the weighted share of household-months with a positive purchase. "Both" is the share buying both products in the month. Quantities are recorded purchases in months with a positive purchase of the product (packs of 20 cigarettes, millilitres of e-liquid), weighted by NielsenIQ projection weights and not winsorised. *N* counts buyer-months.

in which households vape add to the rise. Dual users buy less of each product than single-product users: 8.0 against 9.6 packs, and 26 against 32 mL a month.

Figure 1 plots prices (top row) against participation and quantity per user (bottom rows). Real prices rose modestly: the cigarette price index increases by 13% in 2024 compared to 2019, and the e-cigarette index peaks 10% above it in 2023 before falling back to 3% above. Rising prices did not lower participation or quantities in the raw data. Smoking participation moves between 4.8% and 7.2%, while packs per smoker rise from 6.6 to 11.8. Vaping participation more than doubles, from 0.38% at the beginning of 2019 to 0.83% at the end of 2024. Millilitres per vaper are noisy overtime. The annual mean rises from 24 mL in 2019 to 35 mL in 2024 (Table 4). These co-movements are not price responses, they are contaminated with common demand shocks such as the growing popularity of vaping. The estimation therefore relies on tax-driven price variation across states and controls for local demand shocks (Section 4.1).

Persistence. Behaviour is highly persistent from month to month (Table 5). Among smokers, 73% are followed by another month of smoking only. Among vapers, 62% by another month of vaping only. Direct switching between the two products is rare: 0.3% of smokers move to vaping only and 1.2% of vapers to smoking only in the next month. Dual use is the least stable state. Only 26% of dual users buy both again the next month, and 41% keep cigarettes and drop e-cigarettes.

Table 5: One-month action transitions (%).

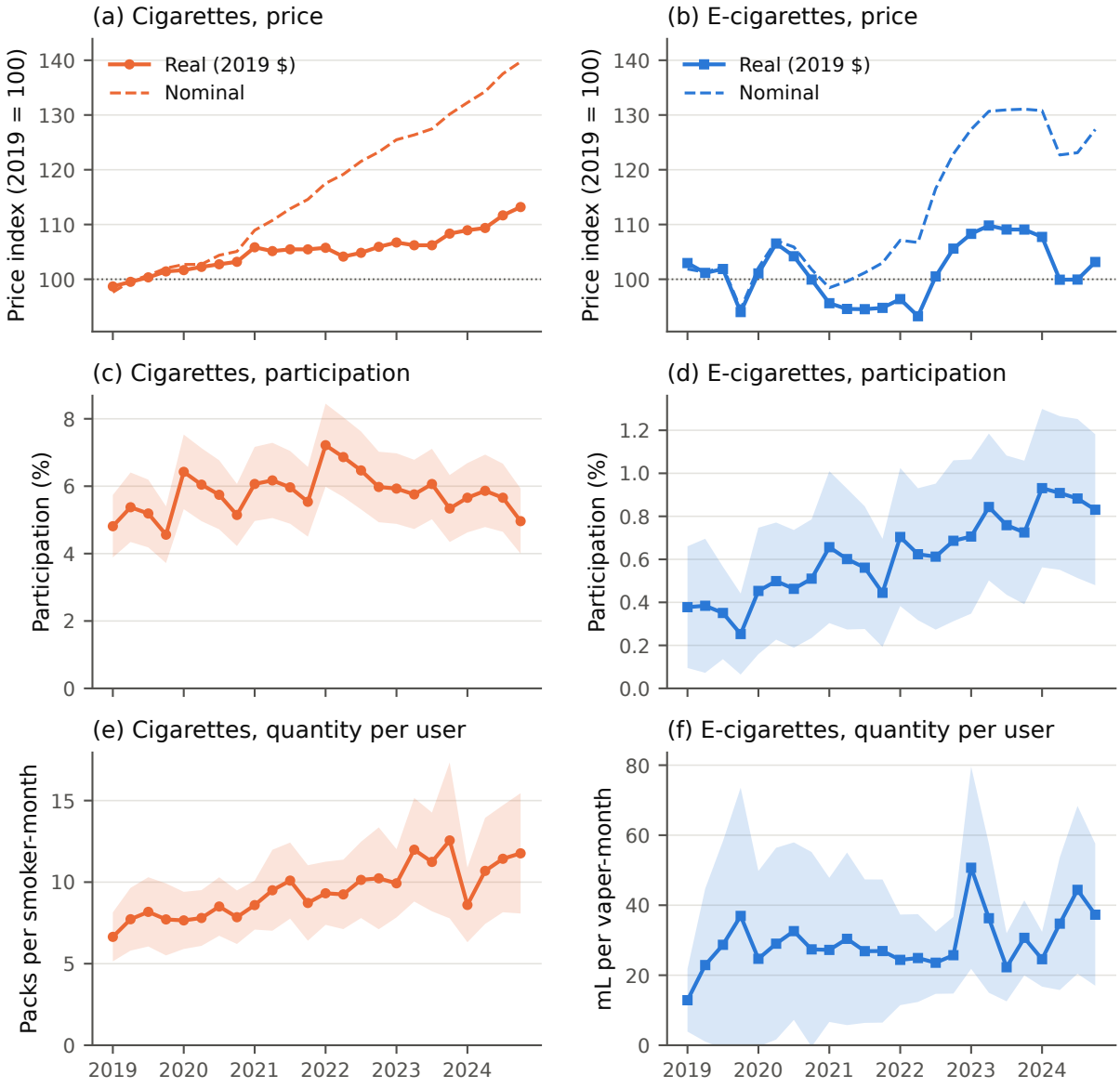
	Action in month t				
Action in month $t-1$	Neither	Smoke only	Vape only	Both	N
Abstain	98.3	1.5	0.2	0.0	163,270
Smoke only	26.1	72.6	0.3	0.9	9,432
Vape only	34.1	1.2	61.8	2.9	934
Both	20.7	41.1	12.3	26.0	234

Notes: Weighted shares of the action in month t given the action in month $t-1$, over pairs of consecutive calendar months of the same household. Rows sum to 100. N counts origin household-months.

6 Estimation Results

Table 6 reports the estimated utility coefficients and Table 7 reports the latent types distribution. In the estimated specification, each quantity shifter Φ^g_{it} depends on the product's own stock Q^g_{it}, and the e-cigarette shifter also on the control function C^v_{ct}. Cigarette market is mature and not subject to major demand shock over the sample

Figure 1: Prices, participation and quantity per user, 2019–2024.



Notes: Quarterly values. Panels (a)–(b): weighted mean county price in 2019 dollars (solid) and in nominal dollars (dashed), indexed to the 2019 average ($6.90 per pack, $5.15 per mL). Panels (c)–(d): weighted share of household-months with a purchase. Panels (e)–(f): weighted mean monthly purchase among buyer-months (packs of 20 cigarettes and millilitres of e-liquid). Shaded bands are 95% confidence intervals clustered by household.

period, and in estimation the coefficients on cigarette control function terms are very close to 0. Therefore, in the final estimation, only control function for e-cigarettes are included. Both prices and stocks enter as z-scores for numerical stability. Standard errors come from the outer product of the household scores.

Addiction and participation. Panel A of Table 6 reports how the two stocks enter the utility of each action. The own-stock terms are positive and precisely estimated for both products (1.18 for smoking and 0.57 for vaping). The squared terms are negative, supporting the diminishing relationship between utility and addiction. Figure 2(b) traces the implied utility over the range of each stock, converted into dollars of monthly spending with the money coefficient. The utility of smoking rises by about \$190 a month as the cigarette stock grows to 3.0 (41% of the ceiling). At the stock ceiling, the utility from addiction falls to negative of −\$218. The utility of vaping follows a similar hump: it rises by about \$170 a month at $Q^v = 1.5$ (42% of the ceiling) and returns to its zero-stock level at 2.9, with wide confidence bands at high vaping stocks.

The cross-stock terms are negative (Figure 2(c)). Smokers are less inclined to switch to vaping only. The cigarette stock lowers the utility of vaping only, and the disutility grows with the stock: it is about \$136 a month for a light smoker, at a quarter of the stock ceiling, and about \$263 at the ceiling, where it is imprecisely estimated. The vaping stock lowers the utility of smoking only, by \$77 a month at a quarter of the vaping-stock ceiling and by \$713 at the ceiling. Dual use is most attractive at moderate cigarette stocks: its linear cigarette-stock term is positive and both squared terms are negative.

Last month's purchase. Panel B reports how last month's purchases enter utility, holding the stocks fixed. The level terms are small and imprecise, except that buying cigarettes last month lowers the utility of smoking only by \$21 a month. The interaction terms are all negative and mostly precise: a household that bought a product last month values additional units this month less. Under smoking only, having bought cigarettes last month is equivalent to a \$1.88 increase in the price of a pack. For vaping only, having bought e-cigarettes last month is equivalent to a \$4.57 increase in the price of a millilitre. These estimates indicate the existence of bulk purchasing and inverntory carried overtime, since the data record purchases rather than consumption.

Quantity shifter. Panel C of Table 6 reports the utility over quantity, $\Phi^g \phi_g(x)$. The cigarette utility shifter rises with the cigarette stock. Because the stock enters $\log \Phi^s$ linearly, each additional unit of stock, which is what a median monthly purchase adds, multiplies Φ^s by the same factor, 1.48: the marginal utility of every pack is 48% higher.

Table 6: Estimated utility parameters.

A. Action utility			
	Smoke only	Vape only	Both
$\tilde{Q}^s$	1.177	−0.823	1.032
	(0.050)	(0.125)	(0.203)
$\tilde{Q}^v$	−0.127	0.573	−0.027
	(0.036)	(0.096)	(0.110)
$(\tilde{Q}^s)^2$	−1.167	0.375	−1.521
	(0.056)	(0.208)	(0.194)
$(\tilde{Q}^v)^2$	−0.350	−0.624	−0.747
	(0.110)	(0.071)	(0.128)
CF, cigarette price innovation λ_{vs}	–	−0.094	
		(0.064)	
CF, e-cigarette price innovation λ_{vv}	–	0.074	
		(0.068)	
B. Last month's purchase			
	Smoke only	Vape only	Both
Bought cigarettes, LP^s (θ^1_{as})	−0.237	−0.037	−0.037
	(0.075)	(0.448)	(0.517)
Bought e-cigarettes, LP^v (θ^1_{av})	−0.081	0.002	−0.058
	(0.601)	(0.558)	(0.502)
$LP^s\times$ quantity (θ^2_{as})	−0.022	−0.035	−0.088
	(0.004)	(0.010)	(0.022)
$LP^v\times$ quantity (θ^2_{av})	−0.094	−0.053	−0.062
	(0.037)	(0.019)	(0.030)
C. Utility of quantity			
	Cigarettes	E-cigarettes	
Shifter constant ψ_{0g}	−2.915	−0.336	
	(0.086)	(0.236)	
Shifter, own stock ψ_g	0.335	0.094	
	(0.012)	(0.013)	
Shifter, CF b_g	–	−0.058	
		(0.103)	
Curvature at zero stock κ_{0g}	1.352	1.457	
	(0.112)	(0.176)	
Curvature, change to the cap κ_{1g}	−0.151	0.106	
	(0.126)	(0.189)	
Measurement error s.d. σ_g	1.129	1.185	
	(0.013)	(0.126)	
D. Money			
Money coefficient α		1.300	
		(0.070)	

Notes: Maximum likelihood estimates, $\beta = 0.95$. Standard errors in parentheses, from the outer product of the household scores. Stocks enter as z-scores $\tilde{Q}$. Squared terms are standardised squared z-scores. The e-cigarette control function (CF) $C^v_{ct} = \rho_v C^v_{c,t-1} + \lambda_{vs} v^s_{ct} + \lambda_{vv} v^v_{ct}$ enters the utility of vaping only and of dual use with a unit coefficient and the e-cigarette shifter with coefficient b_v. Its persistence is calibrated at $\rho_v = 0.83$. $LP^g_{it} = 1$ if the household bought product g last month. The interactions multiply LP^g by the quantity bought in the action (Equation (1)). Under dual use, each quantity interacts only with its own indicator. The curvature follows Equation (4), with $\bar{Q}_g$ the cap of the stock grid (7.38 for Q^s and 3.51 for Q^v). The money term is monthly spending on the two products divided by its sample standard deviation ($113). Not reported in the table: type-specific action intercepts (Table 7), demographic shifters, year-month and county-group effects by action, and the type-probability logit. Sample size: 183,147 household-months with 6,538 households.

Over the 7.4 units between a new smoker ($Q^s = 0$) and a heavy smoker at the stock ceiling ($Q^s = 7.4$, reached by buying about 16 packs every month), these increases compound to $1.48^{7.4} \approx 18$, so Φ^s is 18 times as large (Figure 2(a)).[10] In money terms, an additional pack is worth \$0.49 to a household with no cigarette stock, \$2.33 at half the ceiling and \$11.14 at the ceiling, against a mean price of \$7.27. At the mean price, a household with no cigarette stock would buy 0.6 packs a month in a static choice, and a household at the stock ceiling 7.2 packs. At high stocks, the attraction toward smoking therefore comes from the quantity margin, while the participation utility in Panel A has already turned down and, beyond $Q^s = 6.0$, is below its zero-stock level. The e-cigarette shifter also rises with the vaping stock, by a factor of 1.53 (53%) per unit of stock, which compounds to a 4.5-fold increase between a new vaper and a vaper at the stock ceiling ($Q^v = 3.5$, reached by buying about 18 mL every month, close to the median purchase). The additional millilitre is worth \$0.99 at no vaping stock and \$3.28 at the ceiling (against a mean price of \$5.22).

Curvature. The curvature parameters exceed one at every stock: κ_s is 1.35 at zero stock and 1.20 at the ceiling, and κ_v is 1.46 and 1.56. Neither change with the stock is significant. In the static case with no dynamic behaviours ($\beta = 0$), the price elasticity is -0.86 for cigarettes at zero stock and -0.84 at the ceiling. The price elasticities for e-cigarettes are -0.72 and -0.65 respectively. These elasticities are conditional on use. For cigarettes, the conditional elasticity is larger in magnitude than a historical benchmark discussed by Pesko et al. (2020, Section 2.1): a total demand elasticity of -0.37, with roughly equal contributions from participation and quantity per smoker. For e-cigarettes, Pesko et al. (2020, Section 2.2) discuss an estimated total sales elasticity of -1.5. That estimate includes both margins and is not directly comparable to the conditional elasticity here. Section 7.2 reports the model's total response and its decomposition. The intensive margin is inelastic in the static case. In the dynamic model a purchase also raises next month's stock (Equation (9)), which adds to the value of buying today through addiction.

Money and the control function. The money coefficient is $\alpha = 1.30$. A dollar of monthly spending is worth 0.0115 utility units. This is used to convert utility gains/costs into monetary term. The two loadings of the e-cigarette control function have opposite signs and are not jointly significant, and its coefficient in the e-cigarette shifter is also insignificant. Once the stocks, last month's purchases and the year-month and county-

[10] The stock enters $\log \Phi^g$ as a z-score, so the factor per unit of stock is $\exp(\psi_g / \mathrm{sd}(Q^g))$, with $\mathrm{sd}(Q^s) = 0.86$ and $\mathrm{sd}(Q^v) = 0.22$. The factor does not depend on the level of the stock. The money values are the marginal utility of the median purchase (five packs, 16.8 mL) divided by the money coefficient, and the desired quantities are static ($\beta = 0$) solutions at the mean county price, for a household that did not buy in the previous month.

group effects are controlled for, local demand shocks correlated with e-cigarette price innovations play little role in this specification.

Table 7: Latent types: prior probabilities and action intercepts.

	Households	τ_0 No prior use	τ_s Prior smoker	τ_v Prior vaper	τ_{sv} Prior dual user
A. Prior type probabilities (%)					
All households	6,538	78.7	3.8	5.0	12.5
Under \$25K	1,608	73.6	5.7	5.8	14.8
\$25–45K	1,664	77.2	2.0	6.1	14.6
\$45–70K	1,642	84.0	2.6	2.0	11.4
\$70–100K	985	84.5	3.1	2.0	10.4
\$100K+	639	79.1	5.0	8.0	7.9
B. Action intercepts $c_{\tau,a}$					
Smoke only		−8.71	−2.59	−10.81	−4.52
		(0.28)	(0.21)	(6.83)	(0.20)
Vape only		−17.44	−8.33	−9.41	−10.86
		(2.25)	(1.28)	(1.25)	(1.28)
Both		−21.05	−7.95	−12.94	−15.30
		(7.31)	(1.68)	(2.50)	(1.65)

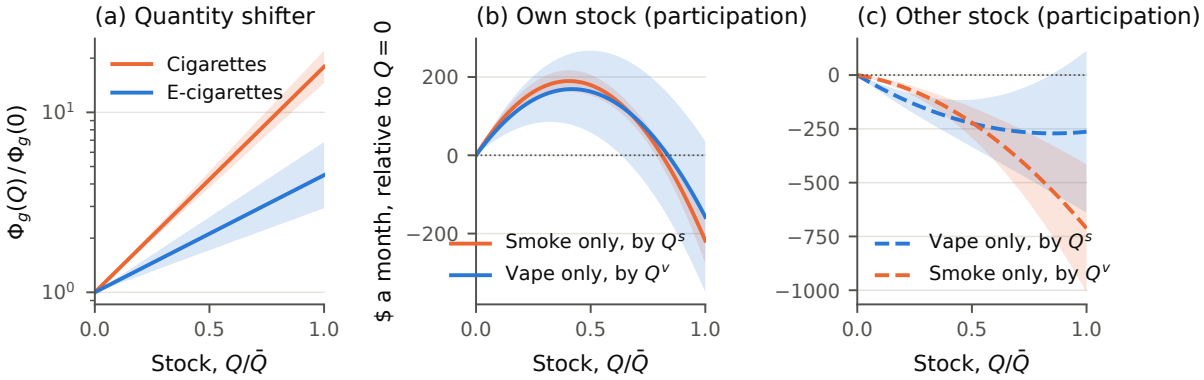
Notes: Panel A: prior probabilities $\pi_\tau(\boldsymbol{z}_i)$ from the estimated type logit, averaged over households with NielsenIQ projection weights. Panel B: standard errors in parentheses. Types with high initial stocks start at the ergodic stock of the 75th-percentile buyer (Section 4.2). The other types start at zero.

Latent types. Table 7 reports the latent types distribution. On average 79% of households have no prior use, 4% are prior smokers, 5% prior vapers and 13% prior dual users. Low-income households are more likely to have high initial stocks: prior dual users make up 15% of households below \$25,000 and only 8% above \$100,000. The highest income group has the most prior vapers (8%). Different types are allowed to have different intercepts for different actions. This is because permanent preference is likely correlated with initial addiction levels. The type intercepts show that the types are more than initial stocks. Prior smokers have the highest utility of smoking, and also the highest utility of vaping only.

6.1 Model Fit

Participation. The fit on the participation margin is mixed. The model predicts abstention in 95.3% of household-months and smoking only in 4.0%, against 93.7% and 5.7% in the data. It under-predicts smoking participation by 26% on average (4.3% of household-months against 5.8%, including dual use). Figure 3 shows the predicted

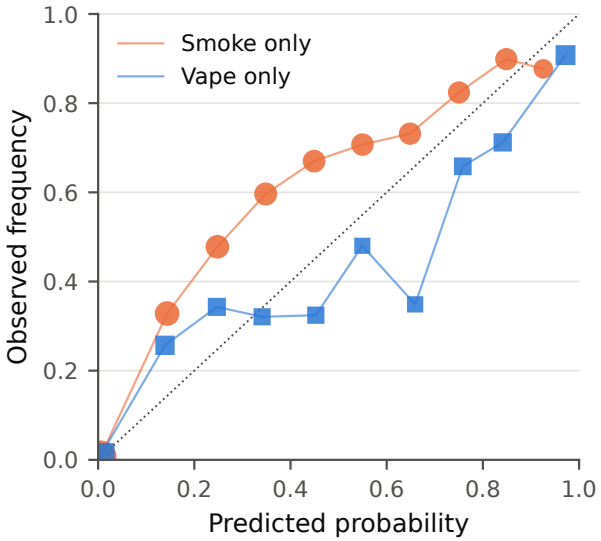
Figure 2: Addiction stocks and utility.



Notes: Panel (a): quantity shifter relative to its value at $Q = 0$ (log scale). Panel (b): participation utility of smoking only by the cigarette stock and of vaping only by the vaping stock, relative to $Q = 0$, in dollars of monthly spending. Panel (c): the same utilities by the other product's stock. Stocks are shown as shares of their ceilings, $\bar{Q}_s = 7.4$ and $\bar{Q}_v = 3.5$. Shaded bands are 95% confidence intervals.

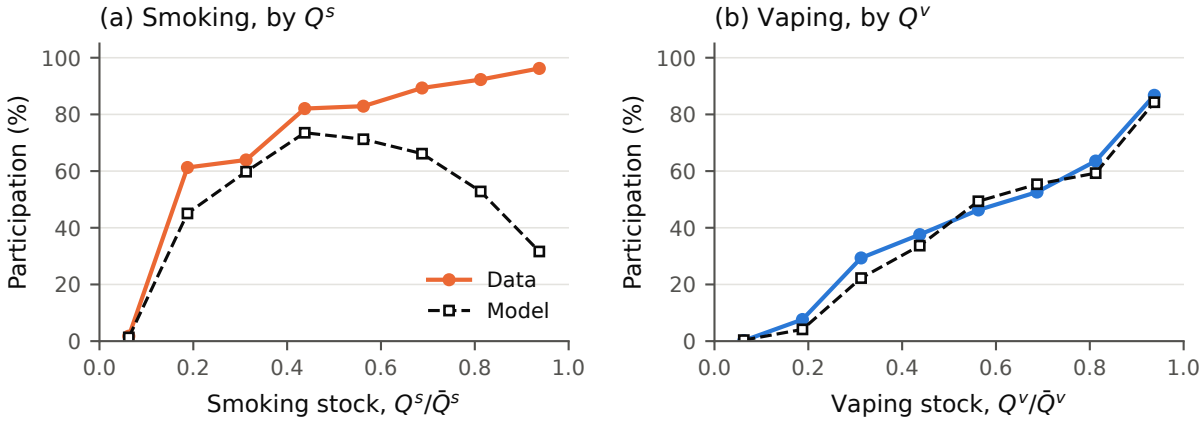
choice probabilities against the observed action frequencies in the data. Figure 4 shows where the shortfall arises. The model matches smoking among households with almost no cigarette stock to mild addiction at $Q^s \approx 3.2$. But the prediction starts to fall short to observed frequency at higher addiction stock levels. For vaping the fit is closer. The model reproduces the increase of vaping participation with the vaping stock (22% against 29% at $Q^v \approx 1.1$, and 84% against 87% in the highest bin), and it predicts overall vaping prevalence of 0.75% against 0.63% in the data. Dual use is over-predicted: 0.32% of all household-months against 0.13% (Appendix Figure 23). By calendar month (Appendix Figure 22), predicted smoking follows the trend of the data but the overall level is lower. The predicted vaping participation tracks the data well in general.

Figure 3: Calibration of predicted action probabilities.



Notes: Observed frequency against predicted probability, by bins of the predicted probability. Marker size increases with the number of household-months.

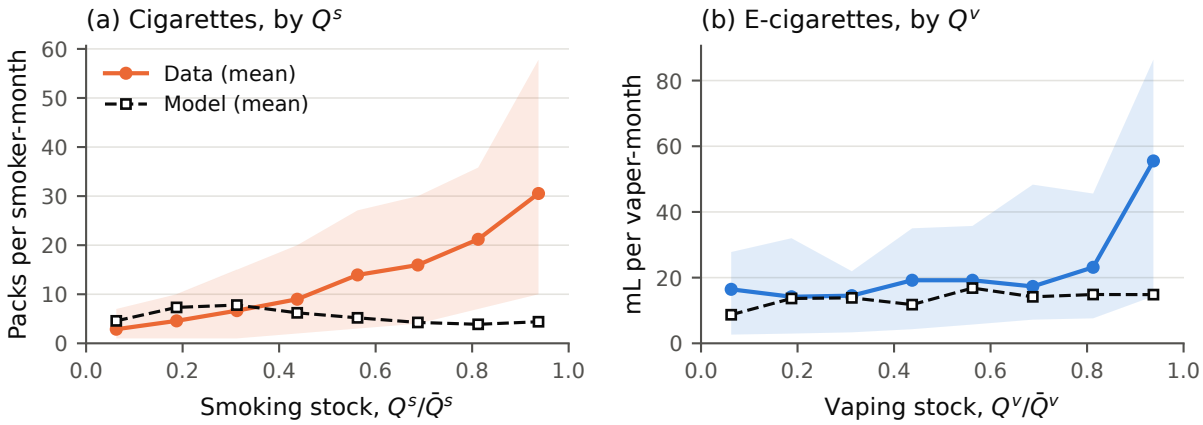
Figure 4: Model fit: participation by stock.



Notes: Share of household-months with a purchase of the product, by the pre-action stock of the same product.

Quantities. The fit on the quantity margin is weaker. Among buyers, the model predicts 5.8 packs and 12.8 mL a month on average, against 9.3 packs and 29.2 mL in the data. Figure 5 shows why: predicted purchases barely change with the stock. For cigarettes, the model over-predicts purchases at low stocks (4.6 against 2.9 packs at $Q^s \approx 0.5$, and 7.3 against 4.6 at $Q^s \approx 1.4$), matches them at $Q^s \approx 2.3$ (7.8 against 6.7), and then predicts 4–6 packs while observed purchases rise to 31 packs in the highest bin. For e-cigarettes, the model matches purchases between $Q^v \approx 0.7$ and 2.4 (12–17 against 14–19 mL), but under-predicts them at near-zero stock (8.7 against 16.4 mL) and in the highest bin (14.8 against 55.5 mL).

Figure 5: Model fit: quantity by stock.



Notes: Monthly purchase among buyer-months, by the pre-action stock of the product. Lines are means. The shaded bands span the 10th to 90th percentiles of the observed purchases. Model predictions are for the chosen action.

Dynamic fit. Figure 24 in Appendix D simulates households forward from a base month along realised prices. It takes the estimation results and each household-month observation as the base period, then simulate the choices and stocks forward. On the

participation margin, the simulation carries the static under-prediction errors forward. In the panel of households observed throughout, smoking falls from 5.8% to 4.6% over ten months, and the simulation from 4.3% to 4.0%. For vaping, the simulated participation rises from 0.7% to 1.1% against a flat 0.6% in the data. On the quantity margin, the simulation starts too low and stays too low. Because smaller cigarette purchases add less to the stock, the simulated cigarette stock falls behind the data, by 15% after ten months. The simulated e-cigarette stock instead ends 33% above the data, as too many simulated households participate in vaping.

7 Policy Experiments

I use the estimated model to study three kinds of tax change: a permanent 1% increase in the price of one product, a prohibitive tax on one product, and the removal of the current excise tax on one product. I first report price elasticities, then trace long-term quantity responses as addiction stocks adjust. I then examine prohibitive taxes and tax removal.

7.1 Design

The first experiment raises the price of one product permanently by 1%. Households expect the increase to persist. I implement it as an added tax in the decision problem, using a surrogate network that takes the tax as an input alongside states and estimated parameters. The experiment specifies the consumer price change directly (i.e. the tax passthrough is assumed to be 1.).

Unless stated otherwise, I evaluate households at their observed states in each sample month. These immediate responses include revised expectations of future prices, but without the subsequent stock adjustment. At zero additional tax, the policy network predicts smoking participation of 3.8% and vaping participation of 1.2%. The estimation network predicts 4.3% and 0.75%, respectively. Each counterfactual is measured against the policy network's own zero-tax baseline. This controls for baseline differences in the comparison, but does not eliminate approximation error.

To include stock adjustment, I simulate the households forward for 24 months from every household-month in the sample under a permanent 1% price increase. Households draw their actions from the model's choice probabilities, and their stocks evolve with the quantities they choose. The first simulated month corresponds to the response at observed stocks, up to simulation error. Later choices also reflect adjusted stocks.

The figures plot each outcome against the pre-action cigarette stock Q^s or vaping stock Q^v. Household-months are grouped into 12 bins on the stock grid. Households within each bin are weighted using a Gaussian density whose bandwidth is one bin width. Bin averages use posterior type probabilities and sampling weights. Total quantity per household is the product of participation and mean quantity conditional on buying. Since households contain one person, this is also quantity per capita. It includes zeros and measures demand across the full sample.

7.2 Price Elasticities by Addiction Stock

I first examine the response to a permanent price increase. Addiction stocks are held at their observed values, while households account for the change in future prices.

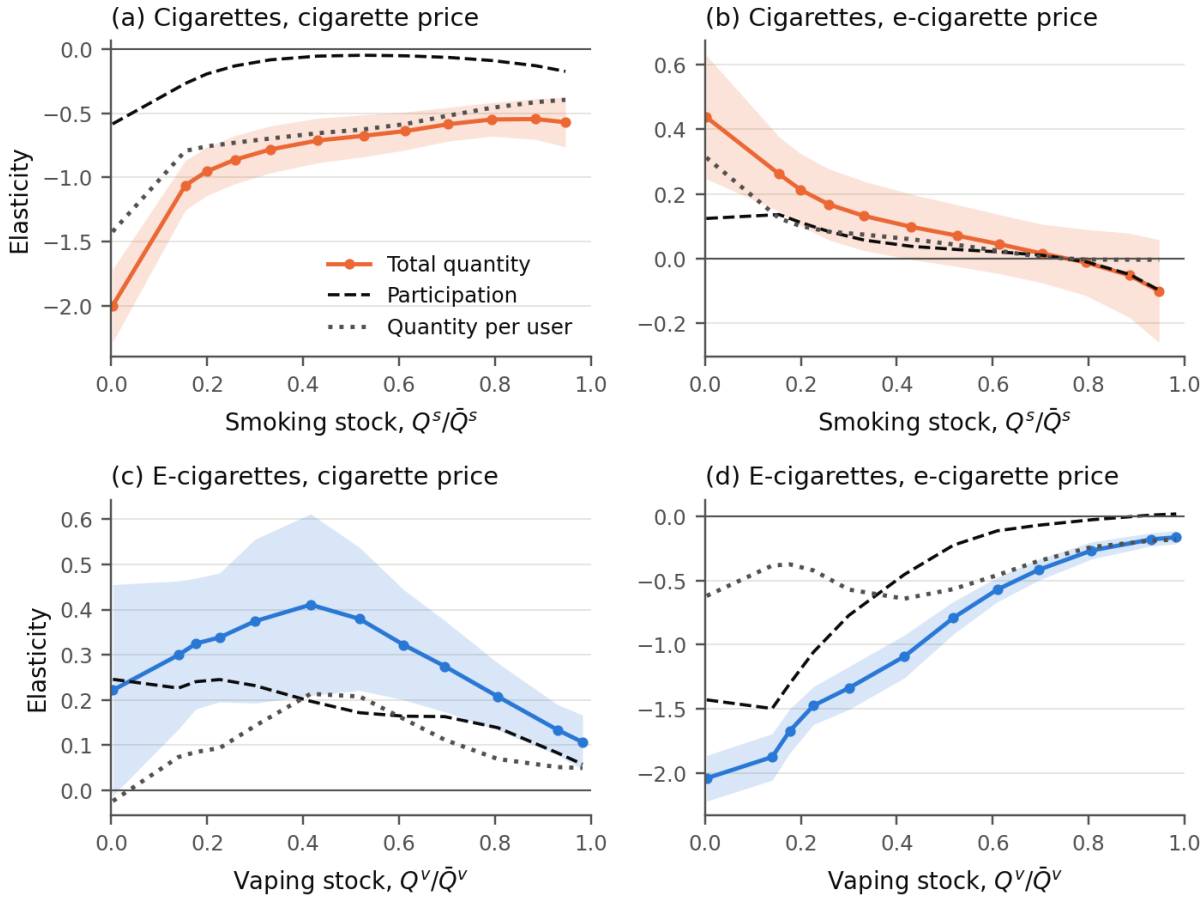
At observed stocks, the aggregate own-price elasticity of total quantity with respect to a permanent price increase is -1.14 for cigarettes and -1.69 for e-cigarettes. For cigarettes, most of the response is on the quantity margin (-0.90, against -0.24 for participation). For e-cigarettes, most of it is on participation (-1.04, against -0.65 for quantity per user). The cigarette elasticity is larger in magnitude than the historical range of -0.3 to -0.5 for total adult cigarette demand reviewed by Chaloupka and Warner (2000) and Pesko et al. (2020, Section 2.1). The cross-price elasticities are positive, so the two products are substitutes, and they are similar in the two directions. A 1% higher e-cigarette price raises cigarette quantity by 0.23%, through both participation (0.08) and quantity per user (0.15). A 1% higher cigarette price raises e-cigarette quantity by 0.24%, almost all of it through participation (0.20).

Own stock. Figure 6 shows the elasticities by the stock of the responding product. The own-price elasticity of cigarette quantity is -2.0 among households with almost no cigarette stock, whose few purchases are small and occasional. The magnitude falls to -0.6 at the stock ceiling (panel (a)). Both margins weaken as smoking stock grows. The participation elasticity falls from -0.58 to below -0.1 beyond $Q^s \approx 3$, and the quantity-per-user elasticity from -1.42 to -0.40. In the model, households with high cigarette stocks are less likely to stop purchasing and their quantities are less sensitive to price. The own-price elasticity of e-cigarette quantity (panel (d)) is -2.0 at almost no vaping stock, mostly through participation (-1.43). It shrinks to -0.2 at the stock ceiling. Addiction to vaping thus makes e-cigarette demand much less sensitive to price on both margins.

The cross-price responses are concentrated among households with little addiction to the responding product. Cigarette quantity responds most to the e-cigarette price among households with almost no cigarette stock (0.44, panel (b)), mostly through quantity per

user (0.31). E-cigarette quantity responds to the cigarette price by 0.36–0.42 at vaping stocks between 0.5 and 1.8, mostly through participation, and by 0.1 at the stock ceiling (panel (c)).

Figure 6: Price elasticities by the stock of the responding product.

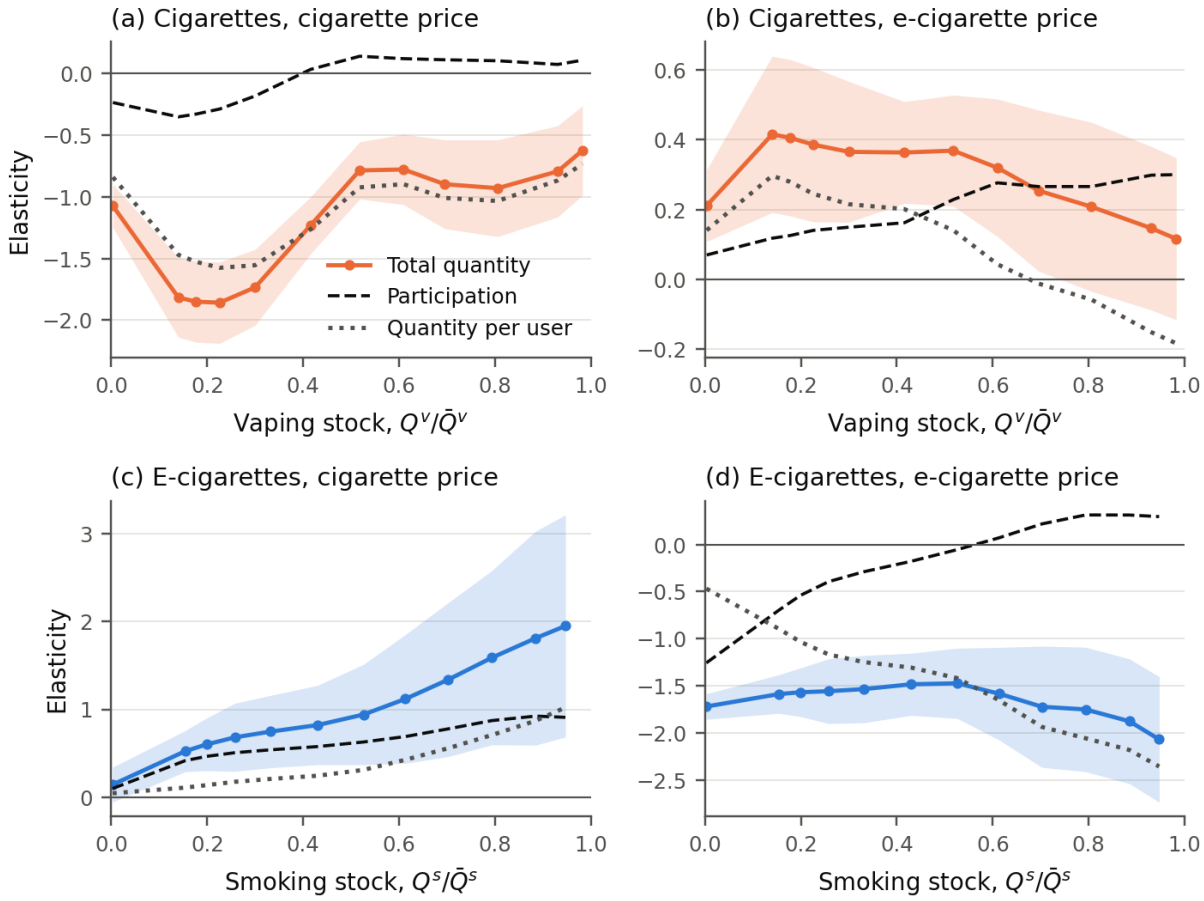


Notes: Elasticities of total quantity per capita with respect to a permanent 1% price increase, by the pre-action stock of the responding product, and their participation and quantity-per-user components. Shaded bands: 95% confidence interval.

Cross-product stock. Figure 7 shows the same elasticities by the stock of the other product. The own-price elasticities show less of a trend with the other product's stock than with their own stock (Figure 6). The cigarette own-price elasticity is −1.07 at almost no vaping stock, around −1.8 at moderate vaping stocks and −0.6 to −0.9 at high vaping stocks (panel (a)). The e-cigarette own-price elasticity lies between −1.5 and −2.1 at every cigarette stock (panel (d)). The cross-price elasticities move in opposite directions with the other product's stock. At high vaping stocks, cigarette demand responds less to e-cigarette prices, i.e. the elasticity of cigarette quantity with respect to the e-cigarette price is 0.36–0.42 at moderate vaping stocks and falls to 0.1 at the top of the grid (panel (b)). At high cigarette stocks, vaping responds more in proportional terms to cigarette prices, i.e. the elasticity of e-cigarette quantity with respect to the

cigarette price rises from 0.14 at almost no cigarette stock to 1.34 at $Q^s \approx 5.2$ and 1.95 at the top of the grid (panel (c)). The band is wide at high cigarette stocks (0.69 to 3.22 at the top), and in the model's zero-tax baseline fewer than 1.5% of household-months at Q^s above 5.8 involve vaping.

Figure 7: Price elasticities by the stock of the other product.



Notes: Elasticities of total quantity per capita with respect to a permanent 1% price increase, by the pre-action stock of the other product, and their participation and quantity-per-user components. Shaded bands: 95% confidence interval.

To summarise, own-price elasticities become smaller in magnitude as the own-product stock rises. At high stocks, most of the remaining response is in quantity per buyer. Cross-price responses also depend on which stock is high. Households with high vaping stocks show little substitution into cigarettes. Households with high cigarette stocks have a large proportional vaping response to cigarette prices.

7.3 Long-term Quantity Responses by Addiction Stock

The above elasticities describe the immediate response to a permanent price increase, before addiction stocks adjust. I next allow stocks to evolve and examine how the responses change over time.

In the dynamic simulations, a permanent 1% increase in the cigarette price lowers cigarette quantity per capita by 1.1% in the first month, and the response stays at about 1.2% over the following two years. Its composition changes as the stocks adjust: the fall in cigarette participation grows from 0.24% in the first month to 0.6% after eight months, and the fall in quantity per user shrinks correspondingly. E-cigarette quantity rises by 0.2–0.3% throughout. A permanent 1% increase in the e-cigarette price lowers e-cigarette quantity by 1.7% in the first month, by about 1.2% after a year, and it raises cigarette quantity by 0.2–0.3% throughout.

Figure 8 shows the change in total quantity per capita after 24 months by the stock at the start of the simulation, with the participation and quantity-per-user components. The own-price responses weaken with the own stock. Cigarette quantity falls by 2.3% among households that start with almost no cigarette stock and by 0.4% at the top of the grid (panel (a)). At low stocks most of the fall is on participation, whereas at high stocks almost all of it is on quantity per user.

E-cigarette quantity falls by 1.3% among households with almost no vaping stock and by 0.3% at high vaping stocks, again with participation dominating at low stocks and quantity per user at high stocks (panel (d)). The cross-price responses are small and positive at every stock. Cigarette quantity rises by 0.1–0.4% when e-cigarettes become more expensive, mostly through participation (panel (b)). E-cigarette quantity rises by 0.3–0.4% when cigarettes become more expensive among households with little vaping stock, but households with high vaping stocks barely change their purchases (panel (c)).

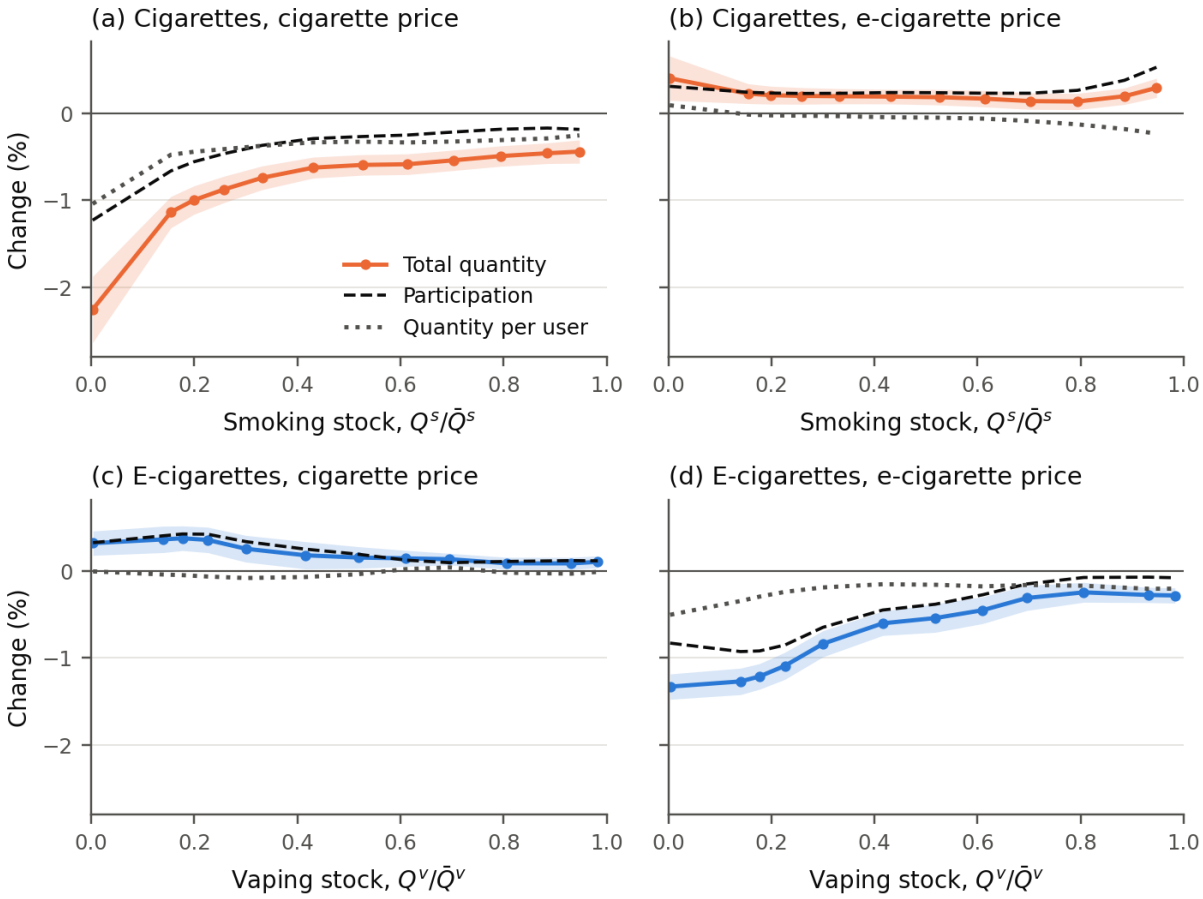
7.4 Prohibitive Taxes and Tax Removal

I consider two larger interventions: adding $999 to the unit price of one product, and subtracting its average excise tax. The first is a prohibitive-tax experiment. It retains the taxed product in the choice set but makes it very expensive. It mimics but is not equivalent to a ban. The second experiment removes the average excised tax.

These prices lie outside the policy network's training range. I therefore fine-tune the network for each scenario, holding the estimated structural parameters fixed. A fine-tune with no tax change serves as a placebo. It moves action shares by at most 0.02 percentage points, and participation by 0.1% for cigarettes and 1.3% for e-cigarettes. I compare the policy changes against these placebo changes to account for the fine-tuning noises.

Aggregate changes. Table 8 reports the share of household-months in each action in the model's zero-tax baseline and after each scenario. The baseline is 95.3% abstinence,

Figure 8: Dynamic response to a permanent price increase, by initial stock.



Notes: Change in total quantity per capita 24 months after a permanent 1% price increase, by the stock at the start of the simulation, and its participation and quantity-per-user components.

3.5% smoking only, 0.9% vaping only and 0.3% dual use.

A prohibitive e-cigarette tax nearly eliminates vaping on both margins. Vaping only falls from 0.91% to 0.01% and dual use from 0.27% to 0.02%. Vaping participation falls in total by 97%, and e-cigarette quantity falls by 99.99%. The 1.14 percentage points that leave vaping-ony and dual-use actions split into a 0.26 point rise in smoking only and a 0.88 point rise in abstinence. Table 10 shows where each baseline group goes: 92% of the vaping-only households move to abstinence and 7% to smoking only, whereas 63% of the dual users move to smoking only and 28% to abstinence. Most vapers quit, but a fair share switch to smoking. Most dual users drop vaping and keep smoking. Cigarette participation rises by only 0.4%, but cigarette quantity rises by 10.7%: the dual users who move into smoking only, together with the smokers who already smoked, replace the lost e-cigarettes with more cigarettes (Table 9).

A prohibitive cigarette tax works differently: cigarette purchases nearly disappear, but participation remains positive. Cigarette quantity falls by 99.6%, while smoking only falls from 3.51% to 1.67%. Including dual use, cigarette participation falls by 48%. In the model the participation decision is tied to the addiction stock, while price affects the quantity choice: households with a large stock keep choosing to smoke, but only buy a negligible amount (the remaining smokers buy 0.05 packs a month). This residual participation is a feature of the model, in which the participation utility does not depend on how much is bought, so the participation response to very large taxes should be read with caution. Nearly all of the 1.84 points that leave smoking only go to abstinence, which rises by 1.80 points: half of the smoking-only households stop buying altogether, and the other half keep the action at a negligible quantity (Table 11). Only 2% of them move to a vaping action. Most vapers and dual users stay in their actions. A few vapers move to abstinence and a few dual users to vaping only. Vaping only and dual use rise by only 0.02 points each, yet e-cigarette quantity rises by 6.8%. The rise comes about equally from the small inflow into vaping (participation +3.5%) and from more quantity per user (+3.0%), the latter driven by dual users, whose mean vaping quantity rises when cigarettes become prohibitively expensive.

Removing the $1.84 cigarette excise does not change vaping: the vaping shares and the millilitres per vaper do not move. It draws smokers from abstinence, raising smoking only from 3.51% to 3.86%, and the smokers buy more: cigarette quantity rises by 34.6%, while quantity per user rises by 23.4%. Removing the $0.31 per millilitre e-cigarette excise does not change smoking: smoking only stays at 3.5% and packs per smoker are unchanged. It raises vaping only from 0.91% to 0.97% and dual use from 0.27% to 0.29%, a 6.4% rise in vaping participation, and e-cigarette quantity by 9.1%.

Table 8: Action shares before and after the prohibitive taxes and the tax removals.

		After			
Action	Baseline	E-cig. tax	Cig. tax	No cig. tax	No e-cig. tax
None	95.31	96.20	97.12	94.98	95.25
Smoking only	3.51	3.77	1.67	3.86	3.50
Vaping only	0.91	0.01	0.93	0.89	0.96
Both	0.27	0.02	0.29	0.27	0.29

Notes: Percent of household-months in each action, averaged over the 25 tax networks (seeds). Across-seed standard errors are at most 0.02 points. Baseline = the tax network at zero tax. The prohibitive taxes add \$999 per pack or per millilitre, and the removals subtract the sample-average state excise tax (\$1.84 per pack, \$0.31 per millilitre). The placebo fine-tune (no price change) moves every share by at most 0.02 points.

Quantities by action. Table 9 reports the purchase size of the households in each action, before and after each scenario. The action is the one chosen after the change, so the dual-use row after the prohibitive e-cigarette tax describes the 0.02% of household-months that still choose both, not the dual users of the baseline. Differences in these means can reflect both changes in purchase size and changes in which households choose the action.

Under the prohibitive e-cigarette tax, packs per smoking-only month rise from 7.6 to 8.0 (6%). Together with the inflow of former dual users and vapers into smoking only, this is the 10.7% rise in cigarette quantity. The few households that remain in the vaping actions buy a negligible amount (0.1–0.2 mL): e-cigarette purchases nearly disappear even when participation remains positive.

Under the prohibitive cigarette tax, the remaining smokers buy 0.05 packs a month, while dual users raise their millilitres from 25.5 to 29.9 (18%) and vaping-only households do not change theirs. The increase in quantity per vaper is therefore concentrated in the dual-use action.

Removing the cigarette excise raises packs per smoking-only month by 23% (7.6 to 9.3) and packs per dual-use month by 17%, while the e-cigarette purchase sizes do not move. Removing the e-cigarette excise leaves packs per smoking-only month and millilitres per vaping-only month almost unchanged, but dual users vape more (25.5 to 26.9 mL, +6%) and smoke less (3.35 to 3.19 packs, −5%): this pattern is consistent with substitution towards e-cigarettes within the dual-use action. Dual use is a small group (0.27% of household-months) and its two cells are the noisiest in the table (the placebo moves them by up to 2%), so the size of this substitution is less precise than the other entries.

Table 9: Purchase size by action before and after the prohibitive taxes and the tax removals.

			After			
Product	Action	Baseline	E-cig. tax	Cig. tax	No cig. tax	No e-cig. tax
Cigarettes (packs)	Smoking only	7.56	8.00	0.05	9.31	7.57
	Both	3.35	8.55	0.05	3.92	3.19
E-cigarettes (mL)	Vaping only	39.9	0.16	40.1	39.7	40.6
	Both	25.5	0.10	29.9	24.8	26.9

Notes: Mean purchase size per month among the household-months choosing the action (packs of cigarettes, millilitres of e-liquid), averaged over the 25 tax networks (seeds). Baseline = the tax network at zero tax. Scenarios are as in Table 8. The placebo fine-tune moves the smoking-only and vaping-only cells by less than 1%, and the dual-use cells by less than 2%.

Action transitions. Tables 10 and 11 show, for each baseline action, how its household-months are distributed over the four actions after the prohibitive tax. Each row sums to 100. The model gives each household a probability over the actions before and after the change rather than an individual transition, so the rows are built by moving, for each household, only the probability mass that leaves an action to the actions that gain mass. Under a prohibitive tax this is close to literal, because the taxed actions lose almost all of their mass. Under the prohibitive e-cigarette tax, most vapers switch to abstinence (92%) but a fair share switch to smoking (7%), while most dual users drop vaping and keep smoking (63% move to smoking only), compensating with more cigarettes (Table 9), while abstainers and smokers stay where they are. Under the prohibitive cigarette tax, abstainers and vapers stay where they are, and almost all smokers stop consuming cigarettes: half of the smoking-only households move to abstinence and the other half keep the action at a negligible quantity, while 2% move to a vaping action. A few vapers (5%) move to abstinence and a few dual users (2.5%) to vaping only.

Table 10: Action transitions under the prohibitive e-cigarette tax.

		Action after the tax (% of the row)			
Baseline action	Share (%)	None	Smoking only	Vaping only	Both
None	95.31	99.9	0.1	0.0	0.0
Smoking only	3.51	1.6	98.4	0.0	0.0
Vaping only	0.91	91.6	7.3	1.1	0.0
Both	0.27	28.0	63.4	0.0	8.6

Notes: Each row distributes the household-months in a baseline action over the actions chosen after a \$999 tax per millilitre of e-liquid, in percent of the row. Share = the baseline share of the action in the population. Built from the minimal-movement coupling of each household's action probabilities before and after the tax, averaged over the 25 tax networks. Across-seed standard errors are below 0.7 points. Under the placebo fine-tune the off-diagonal entries are at most 6.2 (vaping only to none) and 1.9 (both to none, both to smoking only).

Table 11: Action transitions under the prohibitive cigarette tax.

		Action after the tax (% of the row)			
Baseline action	Share (%)	None	Smoking only	Vaping only	Both
None	95.31	100.0	0.0	0.0	0.0
Smoking only	3.51	50.5	47.4	0.6	1.5
Vaping only	0.91	5.3	0.0	94.6	0.0
Both	0.27	9.1	0.0	2.5	88.4

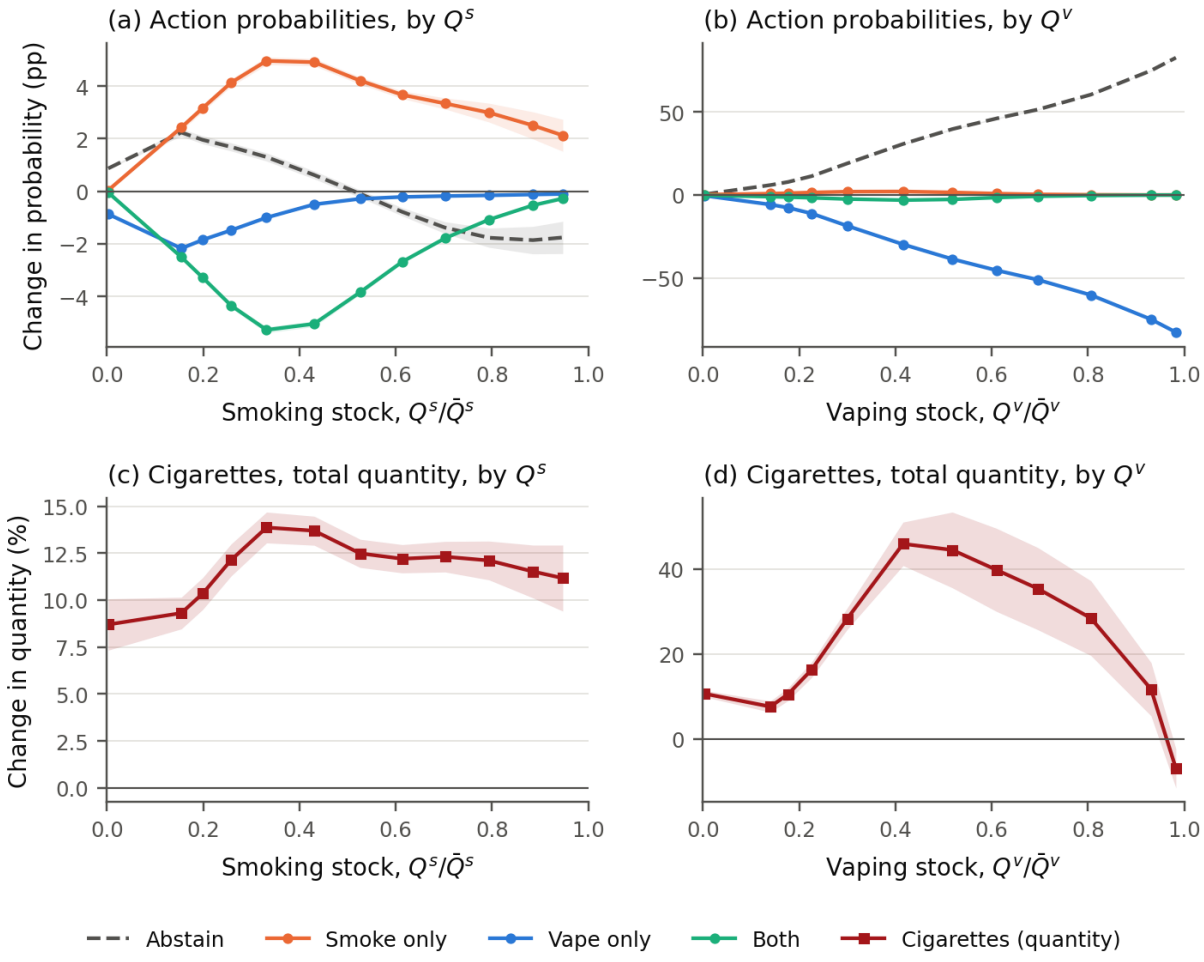
Notes: As Table 10, for a \$999 tax per pack of cigarettes. The smoking-only households that keep the action buy 0.05 packs a month (Table 9). Across-seed standard errors are below 1 point. The movements of vapers and dual users to abstinence are of the size the placebo produces (6.2 and 1.9).

Responses by stock. The remainder of this section looks at each scenario by addiction stock. Each figure has four panels. The top row shows the change in the action probabilities, in percentage points, by the pre-action cigarette stock (panel a) and by the pre-action vaping stock (panel b). The bottom row shows the percent change in total quantity per capita of each product on the same two axes (panels c and d). In the two ban scenarios the banned product's quantity falls by essentially 100% at every stock, so only the other product's quantity is drawn. Bins are kernel-smoothed over the stock grid, the first bin being the households with zero stock. The bands are the across-seed variation of the 25 tax networks. They describe the variation across network fits, not sampling uncertainty.

Prohibitive e-cigarette tax, by stock. Figure 9 shows the changes by stock. Households with high vaping stocks move almost entirely to abstinence. At $Q^v \approx 2.4$, vaping only falls by 51 p.p. and abstinence rises by 51 p.p. Smoking only rises by at most 2.1 p.p. at any vaping stock. The switch to cigarettes comes mostly from dual users. Among households with Q^s between 1 and 4.5, dual use falls by 2.5–5.2 points and smoking only rises by 2.5–5.0 points. Looking the at the quantity response. An e-cigarette ban eliminates almost all e-cigarette demand (quantity drops 100% on vaping). Looking at the change in cigarette consumption, existing smokers all increase cigarette quantiies consumed as a result. The increase are relatively evenly distributed across addiction levels, with new smokers increase smoking quantity by around 9% and heavy smokers increase by 11%. When e-cigarettes are banned, new vapers increase cigarette consumption by 10% while heaviest vapers even slightly decreasae the smoking quantity. Moderately addicted vapers increase the smoking quantity by the most, of nearly 50%.

Prohibitive cigarette tax, by stock. Figure 10 shows the changes by stock. Smoking only falls by 15 p.p. at $Q^s \approx 1.1$ and by 34 p.p. at $Q^s \approx 6.5$, and nearly all of it goes to

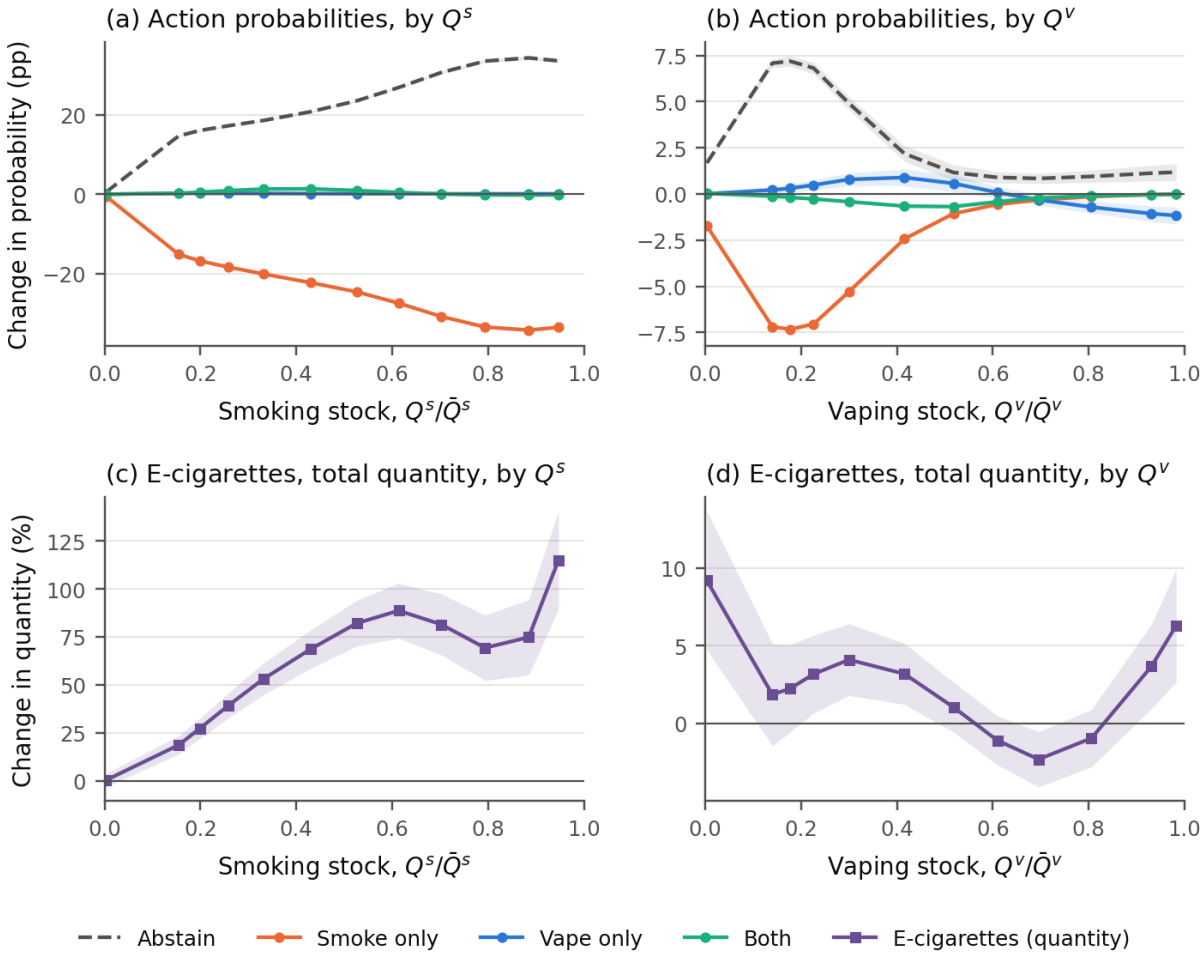
Figure 9: Prohibitive e-cigarette tax.



Notes: Prohibitive tax of $999 per millilitre of e-liquid. Top row: change in action probabilities (percentage points), by the pre-action cigarette stock (panel a) and vaping stock (panel b). Bottom row: percent change in cigarette quantity per capita on the same axes (panels c, d). E-cigarette quantity falls by essentially 100% at every stock and is not drawn. Band: 95% confidence interval.

abstinence. Few of the displaced smokers move to vaping at any stock. Looking then at the quantity response. Smokers reacts to a cigarette ban the strongest, and the response is increasing in their smoking addiction. New smokers barely change vaping consumption while the heaviest smokers increase their vaping consumption by nearly 120%. Vapers react moderately in comparison, with new vapers and heavy vapers react the most by increasing e-cigarette consumption by around 8%. Moderate vapers barely change their vaping quantities.

Figure 10: Prohibitive cigarette tax.

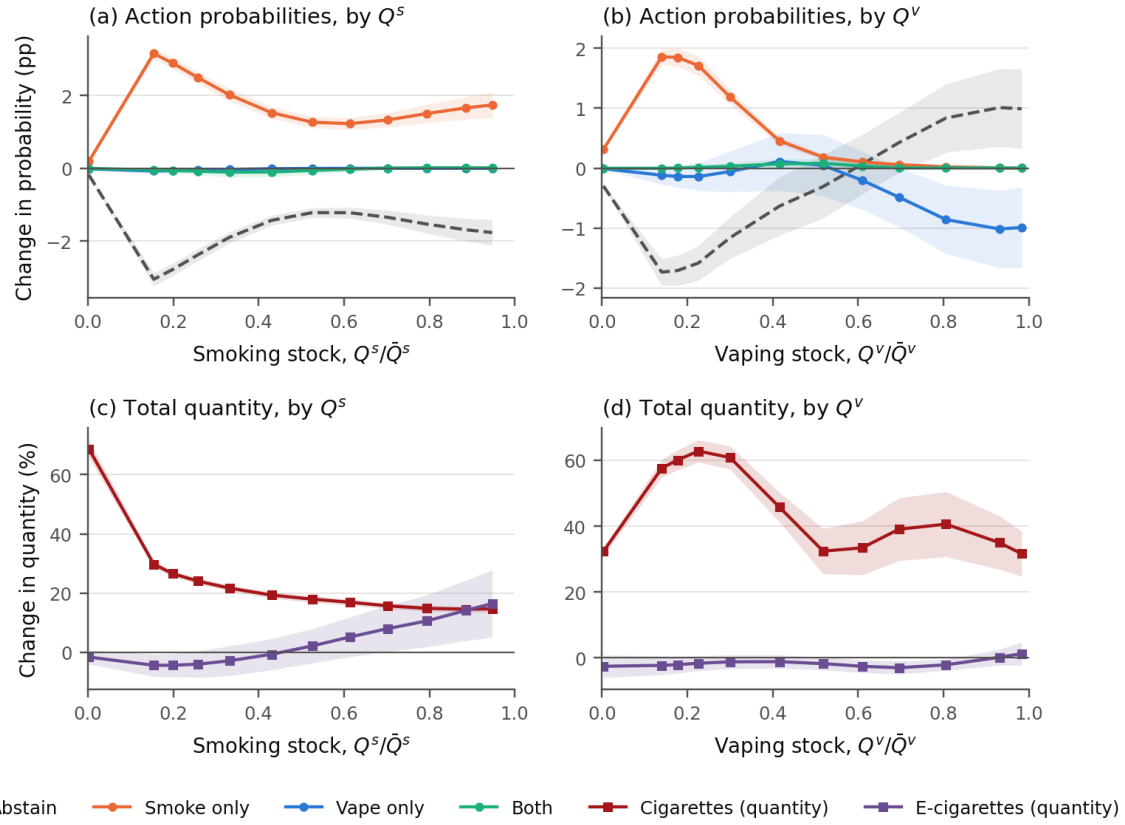


Notes: Prohibitive tax of $999 per pack of cigarettes. Top row: change in action probabilities (percentage points), by the pre-action cigarette stock (panel a) and vaping stock (panel b). Bottom row: percent change in e-cigarette quantity per capita on the same axes (panels c, d). Cigarette quantity falls by essentially 100% at every stock and is not drawn. Band: 95% confidence interval.

Removing the cigarette excise tax, by stock. Figure 11 shows the changes by stock. Smoking only rises by 3.2 p.p. at $Q^s \approx 1.1$ and by 1.7 p.p. at the top of the grid. By vaping stock, smoking only rises by 1.9 points at $Q^v \approx 0.5$. The changes in vaping shares are small and their bands include zero at every stock. These bands describe variation across network fits, rather than sampling uncertainty. In terms of the quantity

responses, cigarette consumption increase for all smokers but the magnitude is decreasing in smoking addiction: new smokers increase their consuption by the most while heavy smokers increase by the least, of aroung 18%. Heavy smokers even increase their vaping consumption. This is likely due to the spill over of built-up addiction from cigarette smoking. Moderately addicted Vapers increase cigarette consumption by the most, of over 60%. New vapers and heavy vapers increase have similar increases in cigarette consumption of around 35%. Vapers' e-cigarette consumptions react little here.

Figure 11: Removal of the cigarette excise tax.

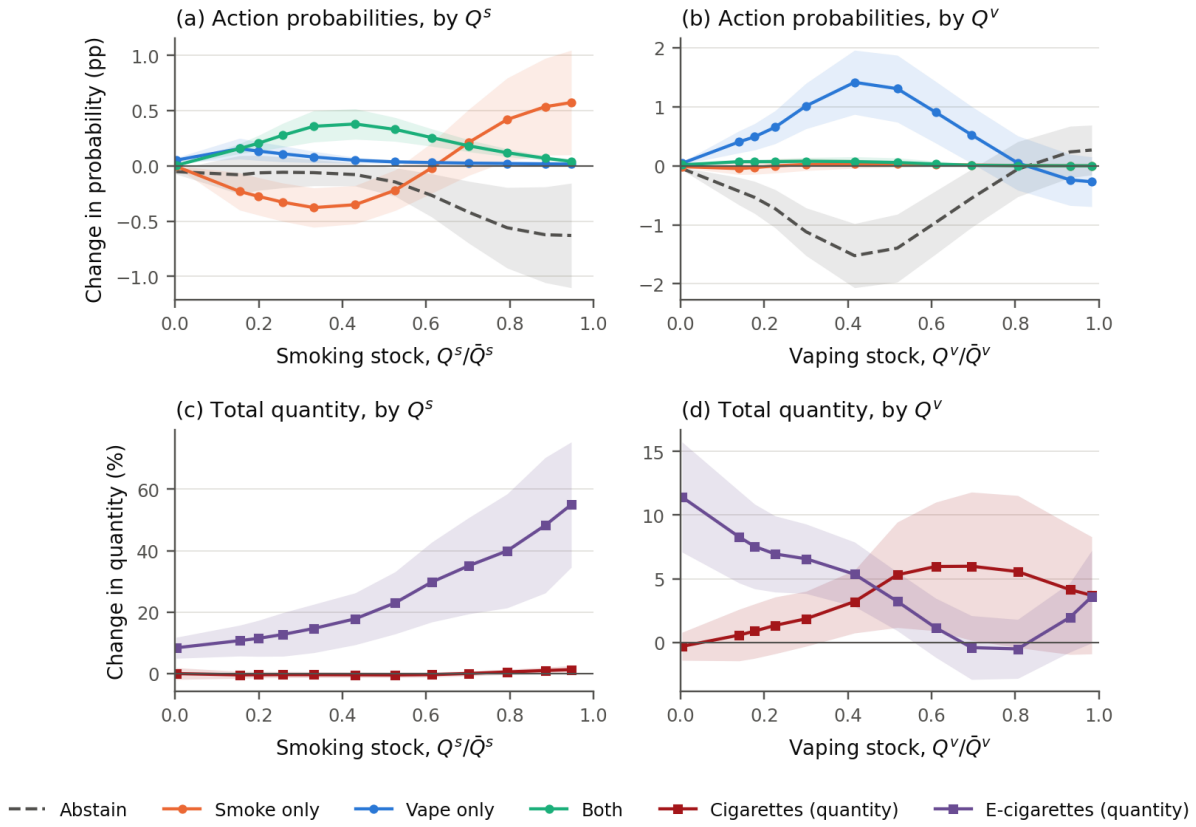


Notes: Removal of the state cigarette excise tax ($1.84 per pack, sample mean). Top row: change in action probabilities (percentage points), by the pre-action cigarette stock (panel a) and vaping stock (panel b). Bottom row: percent change in total quantity per capita of cigarettes and of e-cigarettes on the same axes (panels c, d). Band: 95% confidence interval.

Removing the e-cigarette excise tax, by stock. Figure 12 shows the changes by stock. Vaping only rises by 1.4 p.p. at $Q^v \approx 1.5$. Among households with moderate cigarette stocks (Q^s between 2 and 3.2), smoking only falls by about 0.3 p.p. and dual use rises by about 0.3 p.p.: some smokers add e-cigarettes when they become cheaper, but few switch completely. As for quantity changes, smokers significantly increase their consumption for e-cigarettes, and the response is increasing in smoking addiction. New smokers increase their cigarette quantity demanded by around 8% while heavy smokers increase by nearly 60%. Smokers do not adjust their cigarette consumption at the same

time, indicating a surge in dual-usage. As for vapers, new vapers increase e-cigarette demand by the most, of around 12%. Moderate to heavy vapers do not change their vaping quantity, however, this group of vapers increase their demand for cigarettes. The experiment shows a 5% increase in cigarette quantity demanded for this group, although imprecisely estimated. This indicates a gateway effect story: new vapers are more likely to consume more e-cigarettes and build up addiction, then when they are adequately addicted to nicotine from e-cigarettes, they eventyally increase their cigarettes demand.

Figure 12: Removal of the e-cigarette excise tax.



Notes: Removal of the state e-cigarette excise tax ($0.31 per millilitre, sample mean). Top row: change in action probabilities (percentage points), by the pre-action cigarette stock (panel a) and vaping stock (panel b). Bottom row: percent change in total quantity per capita of cigarettes and of e-cigarettes on the same axes (panels c, d). Band: 95% confidence interval.

Summary. Cigarettes and e-cigarettes are substitutes in the estimated model, with immediate cross-price elasticities of 0.23 and 0.24. Own-price responses weaken with the own-product stock, and the remaining response at high stocks comes mainly through quantity per buyer. Similar patterns remain after two years of stock adjustment. Cross-price elasticities also differ by addiction profile. The prohibitive-tax experiments suggest further differences between existing smokers and exclusive vapers. Their quantitative interpretation requires care because taxed actions remain available and the quantity fit is weak.

8 Discussion and Conclusion

This paper develops and estimates a dynamic discrete–continuous model of cigarette and e-cigarette demand in which two product-specific addiction stocks shift both the utility of each action and the marginal utility of quantity. Unobserved pre-panel addiction is represented by latent types, estimated jointly with the preferences by maximum likelihood, using a surrogate neural network that takes the parameters as inputs.

Addiction raises the utility of smoking and of vaping up to moderate stocks, and for cigarettes it acts mainly on the quantity margin at high stocks: the quantity shifter of a smoker at the stock ceiling is 18 times that of a new smoker. Each product's stock lowers the utility of using the other product alone, and a purchase quantity last month lowers the value of buying more this month. The model fits vaping well but under-predicts smoking among heavy smokers.

In the counterfactual experiments, cigarettes and e-cigarettes are substitutes, with similar cross-price elasticities in the two directions (0.23 and 0.24). The responses depend on addiction: own-price responses weaken as the stocks grow, substitution into vaping comes from heavily addicted smokers, and substitution into cigarettes coming from households with little cigarette stock or a moderate vaping stock. Removing e-cigarettes from the market would send most vapers to abstention and a fair share of them to smoking, while most dual users would keep smoking and buy more cigarettes, raising cigarette consumption by about 11%, whereas removing cigarettes would make almost all smokers stop buying cigarettes, leave most vapers vaping, and raise vaping participation by only about 3.5%. Higher e-cigarette taxes therefore carry a cost in cigarette consumption, borne mainly through dual users and existing smokers.

The results come with limitations. The retention rates are calibrated at $\delta_s = 0.6$ and $\delta_v = 0.7$, which imply habit half-lives of about 1.4 and 1.9 months, and results reported by stock are on a scale that depends on this calibration. A complete estimation will also pick the value of δ based on likelihood. The fit at high cigarette stocks is weak, indicating the model misbehaves at higher smoking stocks.

Next steps. The most immediate extension is to compute welfare measures and the implied optimal cigarette and e-cigarette taxes, which would turn the substitution patterns documented above into a normative prescription. The agenda also includes counterfactuals at addiction specifictax leevels (the optimal tax design). A structural profile-likelihood sweep over the retention rates (δ_s, δ_v) is needed to replace the auxiliary-regression calibration used here.

References

Aguirregabiria, V. and Mira, P. (2002). Swapping the nested fixed point algorithm: A class of estimators for discrete Markov decision models. *Econometrica*, 70(4):1519–1543.

Aizcorbe, A., Corrado, C., and Doms, M. (2003). When do matched-model and hedonic techniques yield similar measures? Working Paper 2003-14, Federal Reserve Bank of San Francisco.

Allcott, H. and Rafkin, C. (2022). Optimal regulation of e-cigarettes: Theory and evidence. *American Economic Journal: Economic Policy*, 14(4):1–50.

Andersson, L., Bask, M., and Melkersson, M. (2006). Economic man and the consumption of addictive goods: The case of two goods. *Substance Use and Misuse*, 41:453–466.

Arcidiacono, P. and Jones, J. B. (2003). Finite mixture distributions, sequential likelihood and the EM algorithm. *Econometrica*, 71(3):933–946.

Arcidiacono, P. and Miller, R. A. (2011). Conditional choice probability estimation of dynamic discrete choice models with unobserved heterogeneity. *Econometrica*, 79(6):1823–1867.

Arcidiacono, P., Sieg, H., and Sloan, F. (2007). Living rationally under the volcano? An empirical analysis of heavy drinking and smoking. *International Economic Review*, 48(1):37–65.

Arrazola, R. A., Husten, C. G., Cornelius, M. E., and Armour, B. S. (2025). Notes from the field: Tobacco product use among adults — United States, 2017–2023. *MMWR Morbidity and Mortality Weekly Report*, 74(7):118–121.

Azinovic, M., Gaegauf, L., and Scheidegger, S. (2022). Deep equilibrium nets. *International Economic Review*, 63(4):1471–1525.

Bask, M. and Melkersson, M. (2003). Should one use smokeless tobacco in smoking cessation programs? A rational addiction approach. *European Journal of Health Economics*, 4(4):263–270.

Bask, M. and Melkersson, M. (2004). Rationally addicted to drinking and smoking? *Applied Economics*, 36:373–381.

Becker, G. S., Grossman, M., and Murphy, K. M. (1994). An empirical analysis of cigarette addiction. *American Economic Review*, 84(3):396–418.

Becker, G. S. and Murphy, K. M. (1988). A theory of rational addiction. *Journal of Political Economy*, 96(4):675–700.

Bhat, C. R. (2008). The multiple discrete-continuous extreme value (MDCEV) model: Role of utility function parameters, identification considerations, and model extensions. *Transportation Research Part B: Methodological*, 42(3):274–303.

Carroll, C. D. (2006). The method of endogenous gridpoints for solving dynamic stochastic optimization problems. *Economics Letters*, 91(3):312–320.

Chaloupka, F. J. (1991). Rational addictive behavior and cigarette smoking. *Journal of Political Economy*, 99(4):722–742.

Chaloupka, F. J. and Warner, K. E. (2000). The economics of smoking. In Culyer, A. J. and Newhouse, J. P., editors, *Handbook of Health Economics*, volume 1, pages 1539–1627. Elsevier.

Chen, H., Didisheim, A., and Scheidegger, S. (2026). Deep surrogates for finance: With an application to option pricing. *Journal of Financial Economics*, 177:104222.

Chen, J. and Rao, V. R. (2020). A dynamic model of rational addiction with stockpiling and learning: An empirical examination of e-cigarettes. *Management Science*, 66(12):5886–5905.

Cornelius, M. E., Wang, T. W., Jamal, A., Loretan, C. G., and Neff, L. J. (2020). Tobacco product use among adults — United States, 2019. *MMWR Morbidity and Mortality Weekly Report*, 69(46):1736–1742.

Cotti, C. D., Nesson, E. T., Pesko, M. F., and Phillips, S. (2026). Standardising the measurement of e-cigarette tax rates in the USA (2nd edition), 2010–2023. *Tobacco Control*, 35(2):173–178.

Cotti, C. D., Nesson, E. T., Pesko, M. F., Phillips, S., and Tefft, N. W. (2023). Standardising the measurement of e-cigarette taxes in the USA, 2010–2020. *Tobacco Control*, 32(e2):e251–e254.

Cragg, J. G. (1971). Some statistical models for limited dependent variables with application to the demand for durable goods. *Econometrica*, 39(5):829–844.

de Haan, J. and Krsinich, F. (2014). Scanner data and the treatment of quality change in nonrevisable price indexes. *Journal of Business & Economic Statistics*, 32(3):341–358.

Dempster, A. P., Laird, N. M., and Rubin, D. B. (1977). Maximum likelihood from incomplete data via the EM algorithm. *Journal of the Royal Statistical Society: Series B*, 39(1):1–22.

Druedahl, J. and Jørgensen, T. H. (2017). A general endogenous grid method for multi-dimensional models with non-convexities and constraints. *Journal of Economic Dynamics and Control*, 74:87–107.

Dubé, J.-P. (2004). Multiple discreteness and product differentiation: Demand for carbonated soft drinks. *Marketing Science*, 23(1):66–81.

Dubin, J. A. and McFadden, D. L. (1984). An econometric analysis of residential electric appliance holdings and consumption. *Econometrica*, 52(2):345–362.

Erdem, T., Imai, S., and Keane, M. P. (2003). Brand and quantity choice dynamics under price uncertainty. *Quantitative Marketing and Economics*, 1(1):5–64.

Farhi, E. and Gabaix, X. (2020). Optimal taxation with behavioral agents. *American Economic Review*, 110(1):298–336.

Fella, G. (2014). A generalized endogenous grid method for non-smooth and non-concave problems. *Review of Economic Dynamics*, 17(2):329–344.

Friedl, A., Kübler, F., Scheidegger, S., and Usui, T. (2023). Deep uncertainty quantification: With an application to integrated assessment models. Working paper.

Gordon, B. R. and Sun, B. (2015). A dynamic model of rational addiction: Evaluating cigarette taxes. *Marketing Science*, 34(3):452–470.

Gruber, J. and Kőszegi, B. (2001). Is addiction "rational"? Theory and evidence. *Quarterly Journal of Economics*, 116(4):1261–1303.

Gruber, J. and Kőszegi, B. (2004). Tax incidence when individuals are time-inconsistent: The case of cigarette excise taxes. *Journal of Public Economics*, 88(9–10):1959–1987.

Hai, R. and Heckman, J. J. (2026). The causal effects of youth cigarette addiction and education. *Journal of Political Economy*, 134(5):1347–1400.

Hajek, P., Phillips-Waller, A., Przulj, D., Pesola, F., Myers Smith, K., Bisal, N., Li, J., Parrott, S., Sasieni, P., Dawkins, L., Ross, L., Goniewicz, M., Wu, Q., and McRobbie, H. J. (2019). A randomized trial of e-cigarettes versus nicotine-replacement therapy. *New England Journal of Medicine*, 380(7):629–637.

Hanemann, W. M. (1984). Discrete/continuous models of consumer demand. *Econometrica*, 52(3):541–561.

Hendel, I. (1999). Estimating multiple-discrete choice models: An application to computerization returns. *Review of Economic Studies*, 66(2):423–446.

Hendel, I. and Nevo, A. (2006). Measuring the implications of sales and consumer inventory behavior. *Econometrica*, 74(6):1637–1673.

Hotz, V. J. and Miller, R. A. (1993). Conditional choice probabilities and the estimation of dynamic models. *Review of Economic Studies*, 60(3):497–529.

Huang, J., Tauras, J., and Chaloupka, F. J. (2014). The impact of price and tobacco control policies on the demand for electronic nicotine delivery systems. *Tobacco Control*, 23(suppl 3):iii41–iii47.

Iskhakov, F., Jørgensen, T. H., Rust, J., and Schjerning, B. (2017). The endogenous grid method for discrete-continuous dynamic choice models with (or without) taste shocks. *Quantitative Economics*, 8(2):317–365.

Ivancic, L., Diewert, W. E., and Fox, K. J. (2011). Scanner data, time aggregation and the construction of price indexes. *Journal of Econometrics*, 161(1):24–35.

Jones, A. M. (1989). A double-hurdle model of cigarette consumption. *Journal of Applied Econometrics*, 4(1):23–39.

Kase, H., Melosi, L., and Rottner, M. (2022). Estimating nonlinear heterogeneous agents models with neural networks. CEPR Discussion Paper 17391, Centre for Economic Policy Research.

Kim, J., Allenby, G. M., and Rossi, P. E. (2002). Modeling consumer demand for variety. *Marketing Science*, 21(3):229–250.

Lewit, E. M. and Coate, D. (1982). The potential for using excise taxes to reduce smoking. *Journal of Health Economics*, 1(2):121–145.

Maliar, L., Maliar, S., and Winant, P. (2021). Deep learning for solving dynamic economic models. *Journal of Monetary Economics*, 122:76–101.

McNeill, A., Brose, L. S., Calder, R., Hitchman, S. C., Hajek, P., and McRobbie, H. (2015). E-cigarettes: An evidence update. Technical report, Public Health England, London, UK. "the current best estimate is that using e-cigarettes is around 95% less harmful than smoking".

National Center for Health Statistics (2025). National health interview survey, 2024. Public-use data file and documentation. `https://www.cdc.gov/nchs/nhis/documentation/index.html`.

NielsenIQ (2024a). Consumer panel data, 2019–2024. Kilts Center for Marketing Data Center, The University of Chicago Booth School of Business.

NielsenIQ (2024b). Retail scanner data, 2019–2024. Kilts Center for Marketing Data Center, The University of Chicago Booth School of Business.

Norets, A. (2012). Estimation of dynamic discrete choice models using artificial neural network approximations. *Econometric Reviews*, 31(1):84–106.

O'Donoghue, T. and Rabin, M. (2006). Optimal sin taxes. *Journal of Public Economics*, 90(10–11):1825–1849.

Orzechowski and Walker (2024). The tax burden on tobacco: Historical compilation. Redistributed by the Centers for Disease Control and Prevention (CDC) STATE System.

Owusu, D., Huang, J., Weaver, S. R., Pechacek, T. F., Ashley, D. L., Nayak, P., and Eriksen, M. P. (2019). Patterns and trends of dual use of e-cigarettes and cigarettes among US adults, 2015–2018. *Preventive medicine reports*, 16:101009.

Pesko, M. F., Courtemanche, C. J., and Maclean, J. C. (2020). The effects of traditional cigarette and e-cigarette tax rates on adult tobacco product use. *Journal of Risk and Uncertainty*, 60(3):229–258.

Pesko, M. F. and Warman, C. (2022). Re-exploring the early relationship between teenage cigarette and e-cigarette use using price and tax changes. *Health Economics*, 31(1):137–153.

Petrin, A. and Train, K. (2010). A control function approach to endogeneity in consumer choice models. *Journal of Marketing Research*, 47(1):3–13.

Pierani, P. and Tiezzi, S. (2009). Addiction and interaction between alcohol and tobacco consumption. *Empirical Economics*, 37(1):1–23.

Rust, J. (1987). Optimal replacement of GMC bus engines: An empirical model of Harold Zurcher. *Econometrica*, 55(5):999–1033.

Saffer, H., Dench, D. L., Grossman, M., and Dave, D. M. (2020). E-cigarettes and adult smoking: Evidence from Minnesota. *Journal of Risk and Uncertainty*, 60(3):207–228.

Scheidegger, S. (2026). Deep learning for solving and estimating dynamic models in economics and finance. arXiv:2605.14493.

Soneji, S., Barrington-Trimis, J. L., Wills, T. A., Leventhal, A. M., Unger, J. B., Gibson, L. A., Yang, J., Primack, B. A., Andrews, J. A., Miech, R. A., et al. (2017). Association between initial use of e-cigarettes and subsequent cigarette smoking among adolescents and young adults: a systematic review and meta-analysis. *JAMA pediatrics*, 171(8):788–797.

Vahratian, A., Briones, E. M., Jamal, A., and Marynak, K. L. (2025). Electronic cigarette use among adults in the united states, 2019–2023. NCHS Data Brief 524, National Center for Health Statistics, Hyattsville, MD.

Zheng, Y., Zhen, C., Dench, D., and Nonnemaker, J. M. (2017). U.S. demand for tobacco products in a system framework. *Health Economics*, 26(8):1067–1086.

A Data Description

This appendix documents the sample, the construction of the variables, and descriptive patterns that complement Section 5. Unless stated otherwise, statistics are monthly, weighted by the NielsenIQ projection weights, and computed on the estimation sample. Confidence intervals are 95% and clustered by household.

Sample

Table 12 traces the sample from all single-person households observed in 2019–2024 to the estimation sample. The age restriction removes three-quarters of the households. The remaining filters (a missing price, incomplete demographics, or residence in one of the 11 states with 30 or fewer sample households) remove a further 241 households.

Table 12: Construction of the estimation sample.

	Households	Household-months
Single-person households, 2019–2024	27,780	1,100,736
Head of household under 50	6,779	189,845
Price observed, complete demographics, states with > 30 sample households	6,538	183,147

Variable construction

Action indicators. For each household-month, I construct the discrete action $a_{it} \in \{0, s, v, sv\}$ defined in Section 5.3. Smoking and vaping participation dummies are

$$d_{it}^{s} = \mathbf{1}\{\text{total cigarette expenditure}_{it} > 0\}, \qquad d_{it}^{v} = \mathbf{1}\{\text{total vape expenditure}_{it} > 0\},$$

and a_{it} is the combination of (d_{it}^{s}, d_{it}^{v}). Quantities and expenditures are summed within the month before the binary measures are formed. Demographics are essentially time-invariant within the window.

Quantities. Cigarette quantities are measured in *packs* of 20 cigarettes bought in the month. E-cigarette quantities are measured in *millilitres of e-liquid*: the number of packages bought times the total millilitres per package, taken from UPC-level size information. Using millilitres rather than device counts matters because vaping products vary widely in capacity (closed-system pods, disposables, and bottles of refill liquid). Millilitres are not nicotine-equivalent across formats, however, because nicotine strength differs across products.

Prices and expenditures. The price state is the county-month price of Section 5.4. The unit value a household pays, expenditure divided by quantity, is never used as the price, because it depends on the household's own choice of product and store. It appears only in the comparison of Figure 15.

Addiction stocks. The observed stocks Q^s_{it} and Q^v_{it} follow the law of motion of the model, Equation (5), with retention rates $\delta_s = 0.6$ and $\delta_v = 0.7$:

$$Q^g_{it'} = \delta_g^{\Delta}\, Q^g_{it} + \delta_g^{\Delta-1}\chi_g(x^{g,\text{obs}}_{it}), \qquad g \in \{s, v\},$$

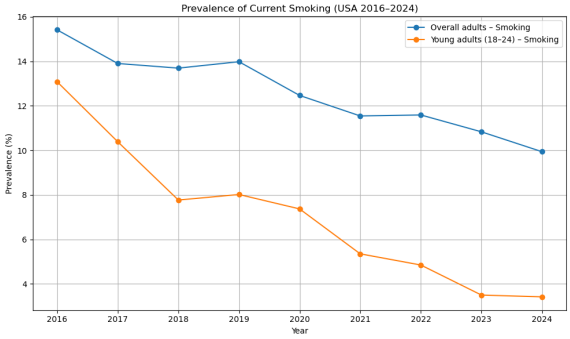
where Q^g_{it} is the pre-action stock of month t, as in the model, t' is the household's next observed month and $\Delta = t' - t$. With $\Delta = 1$ this is Equation (5). Here $x^{g,\text{obs}}$ is the recorded monthly purchase, $\chi_g(0) = 0$, and the median quantity $\bar{x}^g$ in χ_g is computed over buyer-months in the estimation sample (unweighted: 5 packs and 16.8 mL, while the weighted medians in Table 4 are 4 packs and 16 mL). A purchase of the median quantity therefore adds one unit to the stock, and a purchase of three times the median or more adds the cap $K = 3$. The stocks are bounded by $\bar{Q}_s = 7.4$ and $\bar{Q}_v = 3.5$. Households are observed with occasional gaps in the panel, so Δ can exceed one: retention is compounded over the gap, with no imputed purchases during missing months.

Comparison with national prevalence

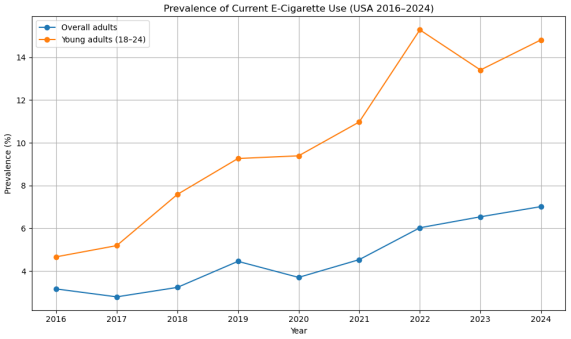
Figure 13 shows the recent trend of cigarette and e-cigarette use among U.S. adults, computed by the author from the National Health Interview Survey (NHIS) Sample Adult public-use files (2016–2024) using the sample-adult final weights (National Center for Health Statistics, 2025). Panels (a) and (b) plot the overall and young-adult (ages 18–24) prevalence of current cigarette smoking and current e-cigarette use, and panels (c) and (d) plot the same series split by sex. Cigarette smoking continues its steady decline, while e-cigarette use has risen rapidly, especially among young adults. Earlier years

contain too few vapers for estimation, while the 2019–2024 window combines substantial vaping prevalence with several state-level tax changes.

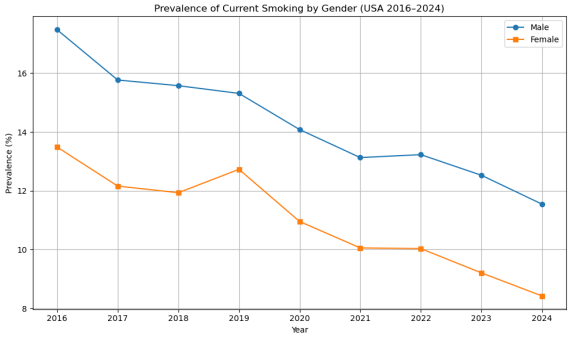
Figure 13: Smoking and e-cigarette use among U.S. adults, 2016–2024.



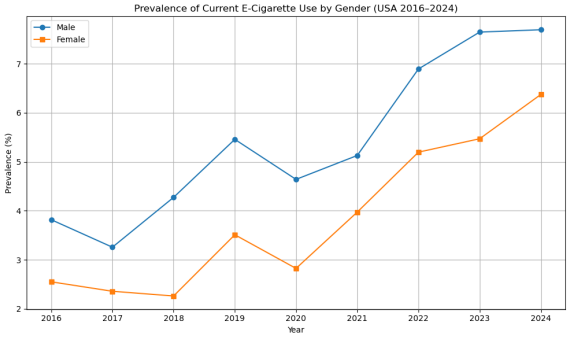
(a) Cigarette smoking: overall vs. young adults (18–24).



(b) E-cigarette use: overall vs. young adults (18–24).



(c) Cigarette smoking: men vs. women.



(d) E-cigarette use: men vs. women.

Notes: Prevalence of current cigarette smoking and e-cigarette use, computed by the author from the National Health Interview Survey (National Center for Health Statistics, 2025). "Current" use follows the standard NHIS definition (now uses every day or some days). Panels (a) and (b) split by age (all adults vs. ages 18–24), and panels (c) and (d) split by sex. See Arrazola et al. (2025) and Vahratian et al. (2025) for related NHIS-based prevalence estimates.

The panel's monthly purchase rates are well below these prevalence figures. Smoking participation is 5.0–6.6% of household-months and vaping participation 0.34–0.89% (Table 4). For adult cigarette smoking, NHIS estimates fell from 14.0% in 2019 (Cornelius et al., 2020) to 11.6% in 2022 and 11.0% in 2023 (Arrazola et al., 2025). For adult e-cigarette use, NHIS estimates were 4.5% in 2019, 3.7% in 2020, 4.5% in 2021, 6.0% in 2022, and 6.5% in 2023 (Vahratian et al., 2025).

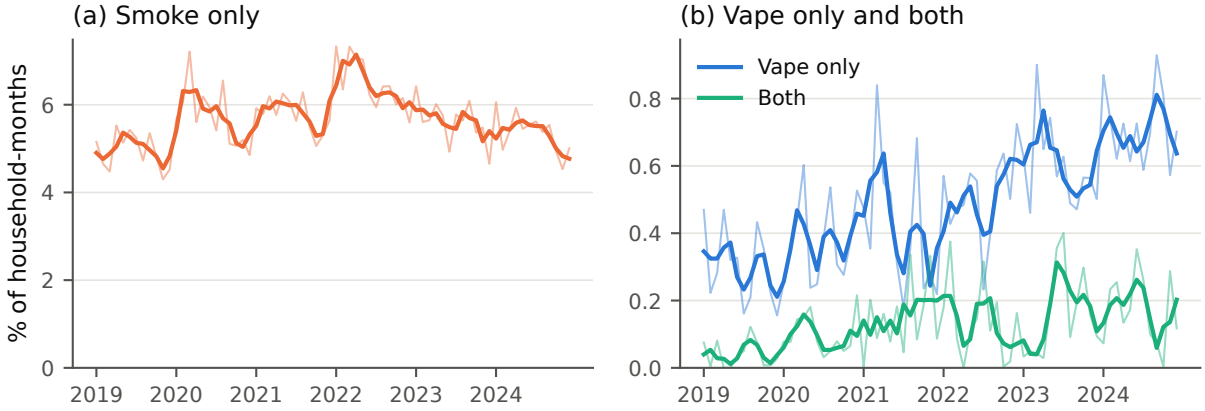
Three features of the data account for most of the gap. First, the two measures differ: a monthly purchase rate counts months with a recorded purchase, while prevalence counts people who use the product on some days. Occasional users who buy rarely, or who use products bought by others, lower the purchase rate. The share of households that ever buy over the panel, 17% for cigarettes and 3.4% for e-cigarettes, is closer to the prevalence concept. Second, panelists record purchases mainly at grocery, mass-merchandise, drug,

club, and dollar stores. Convenience stores, gas stations, tobacco specialty stores, and online vape retailers are under-covered, and they account for a disproportionate share of cigarette and especially e-cigarette sales. Third, single-person households are a non-random subset of the adult population. Appendix C describes a model of the recording margin. The aim of the paper is not to recover population prevalence but to identify within-household responses to price variation, for which detailed, time-stamped purchase records linked to local prices are well suited.

Action shares over time

Figure 14 shows the monthly shares of the three nicotine actions. Smoking-only participation jumps in early 2020 and again in early 2022 and drifts down from mid-2022. Vaping-only participation more than doubles over the panel, and dual use rises from almost zero to about 0.2%. Monthly shares of the two small actions are noisy, because a month contains only 7–32 vaper-months.

Figure 14: Monthly action shares, 2019–2024.



Notes: Weighted share of household-months in each action. Thin lines are monthly values, and thick lines are centred three-month moving averages.

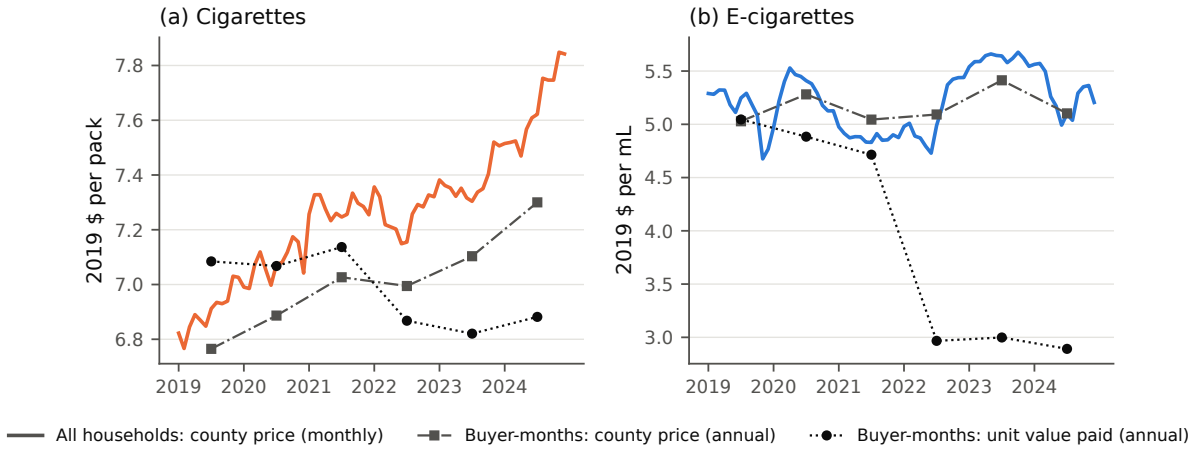
Prices and taxes

Figure 15 shows the real price levels behind the index in Figure 1. Prices differ widely across counties: across sample households, the 10th–90th percentile range of the real cigarette price moves from \$5.15–\$9.49 per pack in January 2019 to \$6.01–\$10.48 in December 2024.

The figure also compares the county price with the unit value buyers actually pay, expenditure divided by quantity in the month, both in 2019 dollars and averaged over the same buyer-months. For cigarettes the two are close: \$6.76 against \$7.08 per pack in

2019 and \$7.30 against \$6.88 in 2024. The county price rises by about 11% within a household between 2019 and 2024, whether or not the household buys cigarettes, while the unit value paid falls slightly. For e-cigarettes the gap is larger. The county price faced by vapers stays near \$5 per millilitre, while the unit value they pay falls from \$5.05 in 2019 to \$2.89 in 2024, most of it between 2021 and 2022 (from \$4.71 to \$2.97). Vapers moved to products that are cheaper per millilitre. The index holds the products fixed, so this shift shows up as more millilitres per vaper-month (Table 4) rather than as a lower price.

Figure 15: Real price levels, 2019–2024.



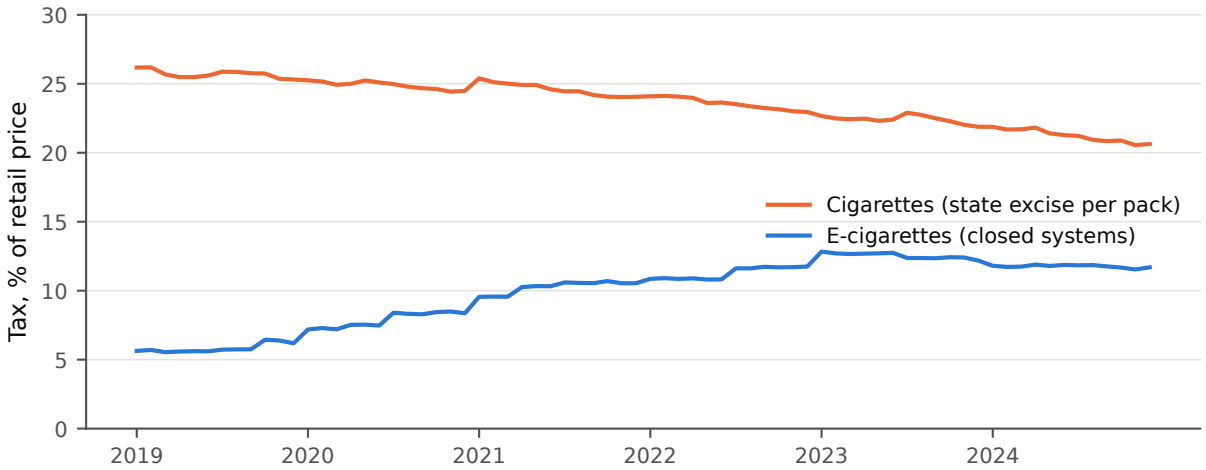
Notes: In 2019 dollars (CPI-U). Solid line: weighted mean county price over all sample households, monthly. Markers: annual means over buyer-months of the county price (squares) and of the unit value paid, expenditure over quantity in the month (circles). All means use NielsenIQ projection weights.

Figure 16 shows state excise taxes as a percentage of the current retail price. The average state cigarette excise falls from \$1.90 to \$1.75 per pack in 2019 dollars, with discrete steps when states raise their rates, and its share of the price falls from 26% to 21%: a tax set in nominal dollars per pack erodes with inflation. E-cigarette taxes rise in steps from 5.6% of the price in January 2019 to 12.8% in January 2023, as states adopt new taxes, and stand at 11.7% at the end of 2024. Over the same period the weighted share of households living in a state that taxes e-cigarettes rises from 26% to 68%.

Quantities

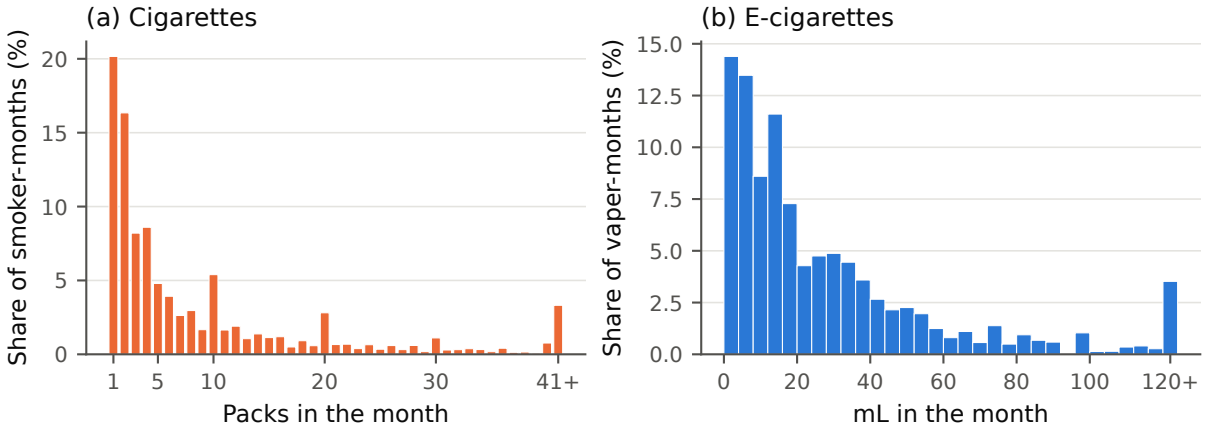
Figure 17 shows the distribution of monthly purchase quantities. One in five smoker-months contains a single pack and one in six two packs. Purchases heap at ten and twenty packs (one and two cartons), and 3% of smoker-months exceed 40 packs. E-cigarette purchases span two orders of magnitude, from 3.5 mL at the 10th percentile to 58 mL at the 90th and 168 mL at the 99th. The heavy right tail of e-cigarette purchases, from households buying refill bottles, is why the mean is well above the median in Table 4.

Figure 16: State excise taxes, 2019–2024.



Notes: Taxes as a percentage of the current retail price, weighted mean over sample households. Cigarettes: state excise per pack over the state's average retail price per pack. E-cigarettes (closed systems: pods and disposables): specific taxes per millilitre over the state's average retail price per millilitre. Ad valorem taxes are the statutory rate on the wholesale price times a wholesale-to-retail ratio of 0.55. Households in states without the tax count as zero.

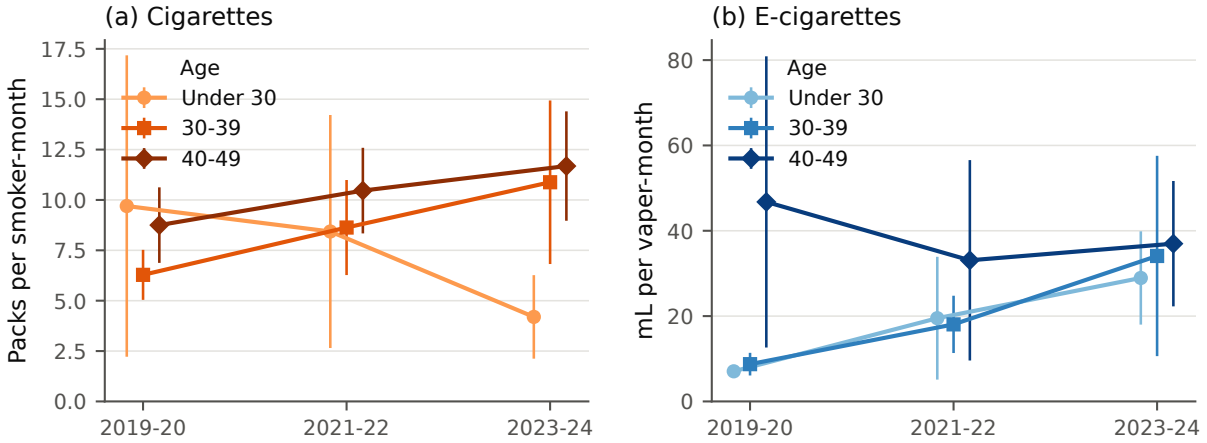
Figure 17: Monthly purchase quantities among buyers.



Notes: Weighted distribution over buyer-months. Panel (a): packs of cigarettes, top-coded at 41. Panel (b): millilitres of e-liquid, 4-mL bins, top-coded at 120 mL.

Figure 18 crosses the two dimensions of Table 4: quantity per user by age group and two-year period. Smokers aged 30–49 buy more over time, while smokers under 30 buy less. Millilitres per vaper-month rise in the two younger groups, from about 8 mL in 2019–20 to around 30 mL in 2023–24, and stay high among vapers aged 40–49. The cells are small: the under-30 cells rest on 20–35 smoking and 4–13 vaping households, and the other vaping cells on 26–66, so these intervals are wide and the under-30 vaping estimates are only indicative.

Figure 18: Purchase quantity by age group and period.



Notes: Monthly purchase quantity per buyer-month by age group and two-year period. Weighted means, with 95% confidence intervals clustered by household.

Table 13 reports two further facts. Panel A shows that dual users buy less of each product than single-product users. Panel B separates the change in quantities over time into within-household change and changes in who buys. The pooled cross-section of cigarette buyer-months shows log quantity rising by 0.27 between 2019 and 2024, but with household fixed effects the change is −0.15. The rise in packs per smoker therefore comes from the composition of smoker-months shifting toward heavier smokers, not from smokers buying more. For e-cigarettes, quantity rises within households too.

Purchase gaps

Figure 19 shows the number of weeks between consecutive purchase weeks of the same household, computed from the weekly panel from which the monthly data are built. A gap counts only if the household is in the sample in every month it spans, so exits from the panel do not create long gaps. For both products most gaps are short: the median gap is one week, and 87% of cigarette gaps and 83% of e-cigarette gaps are four weeks or shorter. A monthly period therefore contains most purchase cycles. Long gaps are more common for e-cigarettes: 8.3% of e-cigarette gaps exceed eight weeks, against 5.6% for cigarettes, consistent with pods and bottles lasting longer than a pack. A month without

Table 13: Purchase quantities by action, and composition versus within-household change.

A. Quantity per buyer-month, by current action				
	Packs (cigarettes)		mL (e-cigarettes)	
	Mean (median)	*N*	Mean (median)	*N*
Single product	9.6 (4.0)	9,897	31.5 (18.0)	971
Both products	8.0 (4.0)	247	26.0 (14.4)	247
B. Change in log quantity per buyer-month relative to 2019				
	Cigarettes		E-cigarettes	
	Pooled	Within HH	Pooled	Within HH
2020	0.09 (0.10)	0.06 (0.07)	0.04 (0.22)	0.52 (0.10)
2021	0.20 (0.11)	0.10 (0.10)	0.13 (0.27)	0.37 (0.13)
2022	0.16 (0.11)	-0.05 (0.09)	0.39 (0.33)	0.12 (0.13)
2023	0.31 (0.12)	0.00 (0.09)	0.60 (0.37)	0.20 (0.13)
2024	0.27 (0.12)	-0.15 (0.10)	0.71 (0.38)	0.48 (0.19)

Notes: Panel A: weighted mean (median) monthly purchase among buyer-months. "Single product" is smoke-only months for packs and vape-only months for mL. Panel B: weighted regressions of log monthly purchase on year dummies over buyer-months. "Within HH" adds household fixed effects. Standard errors clustered by household in parentheses.

an e-cigarette purchase is therefore somewhat weaker evidence of non-use than a month without a cigarette purchase, consistent with the slower calibrated depreciation of the vaping stock (Section 4.5).

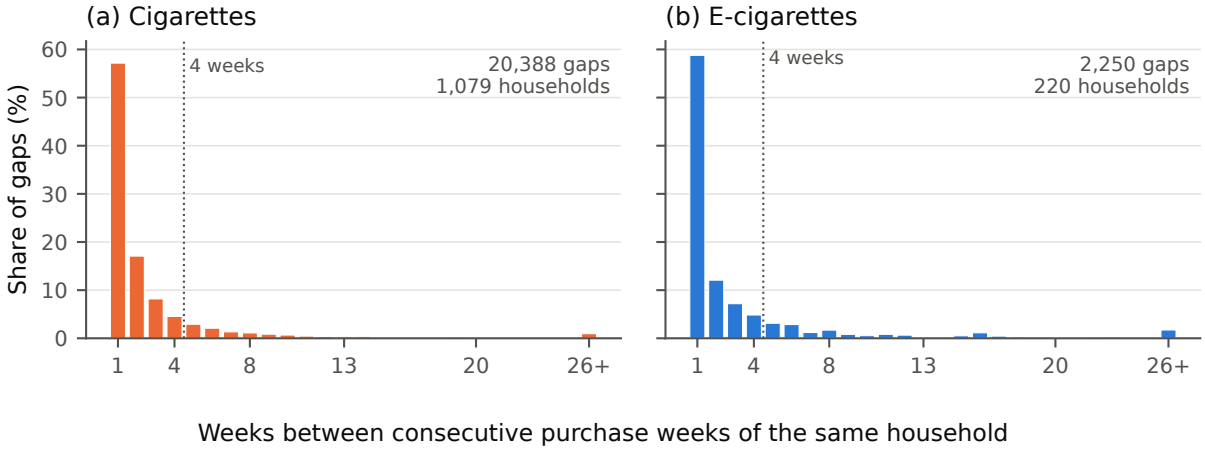
Participation by other demographic groups

Figure 20 extends the age and income gradients of Table 4. Smoking participation falls steeply with education, from 9.8% of household-months without a college degree to 2.1% with a postgraduate degree, and is highest among households not working (13.6%). It is very low among Asian households (0.6%) and similar for men and women. Vaping participation is about a tenth as high, with broadly similar gradients.

Transitions by age group

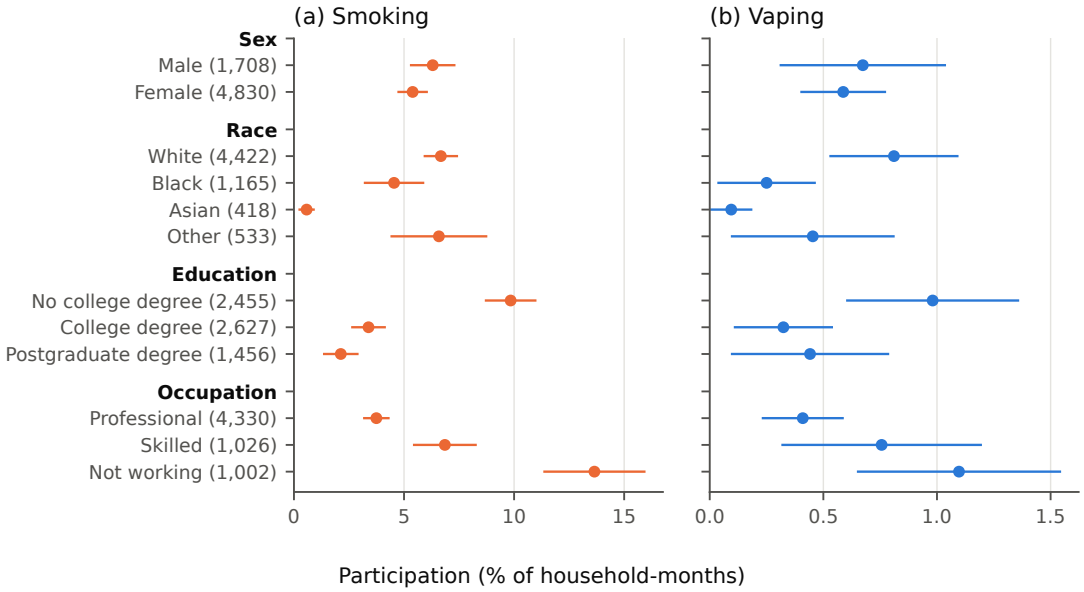
Figure 21 disaggregates the transition matrix of Table 5 by age group. Persistence rises with age in every participation state. Smokers keep smoking in 61% of next months under 30, against 75% at ages 40–49, and vapers keep vaping in 39% against 74%. Younger users mostly exit to non-use: 56% of vaper-months under 30 are followed by a month

Figure 19: Weeks between purchases.



Notes: Weeks between consecutive purchase weeks of the same household, weighted by NielsenIQ projection weights. Gaps of 26 weeks or more are top-coded, and the dotted line marks four weeks.

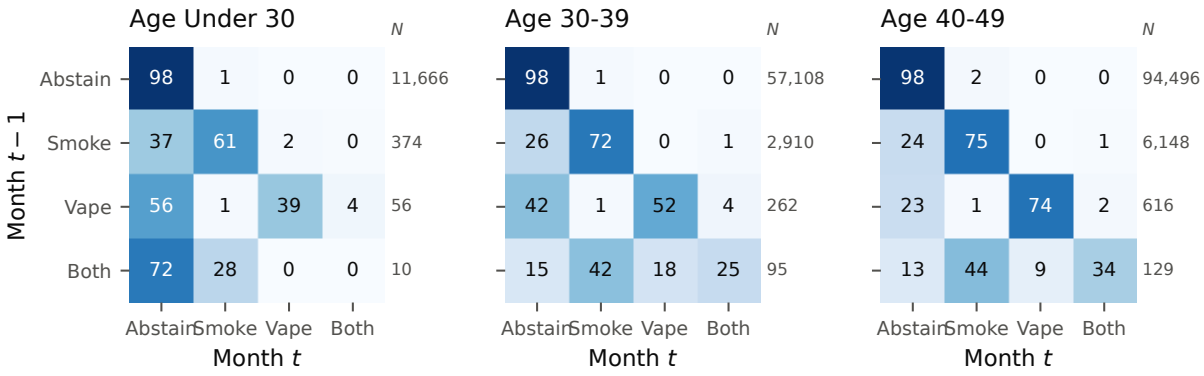
Figure 20: Participation by demographic group.



Notes: Weighted share of household-months with a purchase, with 95% confidence intervals clustered by household. Numbers of households in parentheses. Occupation groups with fewer than 200 households (farmers, and those retired or unable to work) are omitted.

without any purchase, against 23% at 40–49. Rows with few origin months are imprecise. The under-30 dual-use row rests on 10 household-months.

Figure 21: One-month action transitions by age group.



Notes: Percent. Each panel shows the weighted distribution of the action in month t given the action in month $t-1$, over consecutive calendar months. Rows sum to 100. N is the number of origin household-months.

These patterns establish three facts that the structural model must reconcile. First, participation is low, with smoking flat and vaping rising over the panel. Second, prices vary substantially over time and across counties, and taxes contribute much of this variation. Third, both participation and quantity are strongly persistent and build up with past use, while direct switching between the products is rare. Sections 6 and 7 return to these implications.

B Price Index Construction

Retail prices. The RMS data report weekly unit sales and revenue for each UPC in each participating store. For every store-week-UPC cell I take the unit-weighted median shelf price and divide it by the package size, which gives a price per pack of 20 cigarettes or per millilitre of e-liquid. The cells are then collapsed to county-month-UPC sufficient statistics.

Composition-adjusted index. A county's average shelf price mixes the price level with the products its stores carry: a county whose stores stock more cartons and discount brands looks cheap even if every product is priced as elsewhere. Equation (33) separates the two. It is estimated by weighted least squares, with units sold as weights, by alternating projections on the UPC and county-month effects. The dollar index is the cost of a fixed basket at the fitted prices,

$$P_{ct} = e^{\hat{\gamma}_{ct}} \sum_{j} \bar{w}_j e^{\hat{\alpha}_j},$$

where $\bar{w}_j$ is the national unit share of UPC j in 2019. The index is the arithmetic cost of the basket. The exponential of the mean UPC effect would be a geometric index, which understates the basket cost when relative prices are dispersed. The index does not depend on how the UPC and county-month effects are normalised.

Chaining. One set of UPC effects for the whole panel is restrictive for e-cigarettes: relative prices drift and the 2019 basket covers about half of 2024 sales. The regression is therefore estimated on overlapping two-year windows $\{y, y+1\}$, each with its own UPC effects and its own basket (2019 shares in the first window, the window's first-year shares otherwise). The first year takes its values from the first window. Each later year takes the values of the window in which it is the second year, shifted by one constant so that this window agrees, on average over the counties and months of the overlap year and weighted by units, with the values already assigned to that year. Every year-to-year change is thus measured within one window on that window's common products, as in the rolling-window indices of Ivancic et al. (2011). Both products use the chained index. For cigarettes it is almost identical to the fixed-basket index (the standard deviation of the log difference is 0.007). For e-cigarettes the standard deviation is 0.045.

Thin cells. County-months whose index rests on fewer than five UPCs take the state index, built in the same way at the state level. This applies to 3% of sample household-months for cigarettes and 15% for e-cigarettes.

Price faced by households. Every household faces the index of its county, $p^g_{ct} = P^g_{ct}$, deflated by the CPI-U (2019 = 1). Two properties follow. The price does not respond to a household's own purchase in a given month, and it does not depend on the household's choice of brands, pack sizes and stores. These choices are correlated with tastes (heavier smokers tend to buy cheaper brands), so a price built from a household's own unit values would mix its price response with its tastes. Local demand shocks that move prices and purchases together are handled by the control function of Section 4.1.

C Under-Recording of Tobacco Purchases

Recorded purchases in the Homescan panel fall well short of survey use in the same population. Table 14 compares single-person Homescan households with NHIS adults living alone in 2019–2024 (all ages). Recorded annual cigarette purchasing is 59% of NHIS current smoking, recorded monthly purchasing 30% of monthly smoking, and recorded

packs 34% of reported consumption. For e-cigarettes the annual ratio is 17%. Differences in demographic composition explain only 6–14% of the cigarette gap and 24–44% of the e-cigarette gap. The rest lies within demographic cells.

Table 14: Homescan recorded purchases and NHIS current use, single-person households and adults living alone, 2019–2024.

	Homescan	NHIS	Ratio
Any cigarette purchase in the year / current smoker	8.8%	14.9%	0.59
Any e-cigarette purchase in the year / current e-cigarette user	1.0%	5.9%	0.17
Months with a cigarette purchase / smoked in the month	4.4%	14.7%	0.30
Packs per year, purchasers / current smokers	68	203	0.34

A recording model. Let a_{it} be the true action and o_{it} the recorded one. Suppose recording is one-sided: a recorded purchase is a true purchase, but a true purchase of product g is recorded only with probability r_k^g, which varies across demographic cells k. With the purchase month recorded either completely or not at all, the probability of recording o given the true action a is

$$M(o \mid a; r) = \begin{array}{c|cccc} a \backslash o & 0 & s & v & sv \\ \hline 0 & 1 & 0 & 0 & 0 \\ s & 1 - r^s & r^s & 0 & 0 \\ v & 1 - r^v & 0 & r^v & 0 \\ sv & (1 - r^s)(1 - r^v) & 0 & 0 & 1 - (1 - r^s)(1 - r^v) \end{array}$$

and the likelihood of a recorded month replaces the choice probability with $\sum_a M(o_{it} \mid a; r)\, P_\tau(a \mid \boldsymbol{S}_{it}, \boldsymbol{z}_i; \theta)$. Only recorded zeros mix: a month without a recorded purchase is a true abstention or an unrecorded purchase, with weights that depend on the state, so a household with a high addiction stock that records nothing has most likely bought. The rate can be calibrated as the ratio of the Homescan monthly purchase rate to the NHIS monthly use rate in the cell, the latter from the NHIS question on use in the past 30 days. If unrecorded purchases also raise the addiction stock, the stock becomes a hidden state. The likelihood is then a sum over true purchase paths, which can be computed by a forward recursion over a small set of candidate stock values for each household.

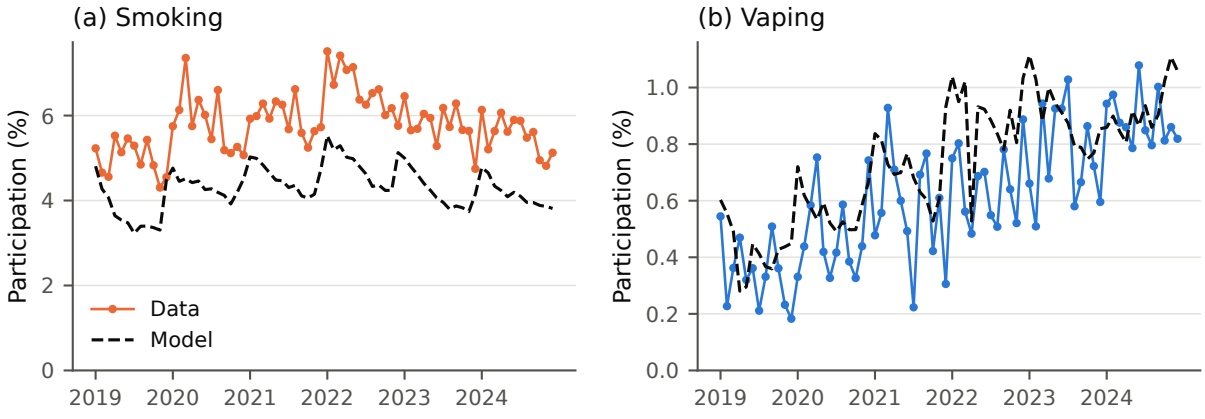
Why the main estimates do not use it. I estimated both versions in exploratory runs with earlier versions of the model. The cigarette margin is estimable: the implied true monthly smoking rate rises from the recorded 5.3% in that exploratory sample to about 9%, recovering roughly half of the gap to NHIS. The e-cigarette margin is not. The calibrated e-cigarette rate is about 0.045, and it does not separate recording from

purchase frequency: pods and bottles last weeks, so a current vaper need not buy every month, and 39% of households that record vaping do so in only one month of a year. At such a rate a recorded zero says almost nothing about vaping, so the vaping probability of the 97% of households that never record a vaping purchase is unrestricted by the data. With 220 vaping households and 0.6% of household-months with a vaping purchase, the estimates degenerated: the addiction-stock effects on vaping turned convex and the implied true vaping rate approached one. Because substitution between the two products is the object of the paper, and the vaping side is the thin part of the data, the main estimates treat recorded purchases as true purchases. The estimated preferences therefore describe recorded purchasing. To the extent that e-cigarette purchases are under-recorded, the vaping intercepts absorb it, and the level of vaping implied by the model should be read with this in mind.

D Additional Model-Fit Diagnostics

Figure 22 compares observed and predicted participation by calendar month. The model follows the time pattern of smoking, including the step up in 2022, but stays 1.2 to 1.8 percentage points below the data in every year. Predicted vaping rises over the panel as in the data, from 0.43% of household-months in 2019 to 0.92% in 2024, against 0.34% and 0.89% observed. Predicted series are smoother than observed shares, especially for vaping, whose monthly shares rest on few households.

Figure 22: Model fit by calendar month.

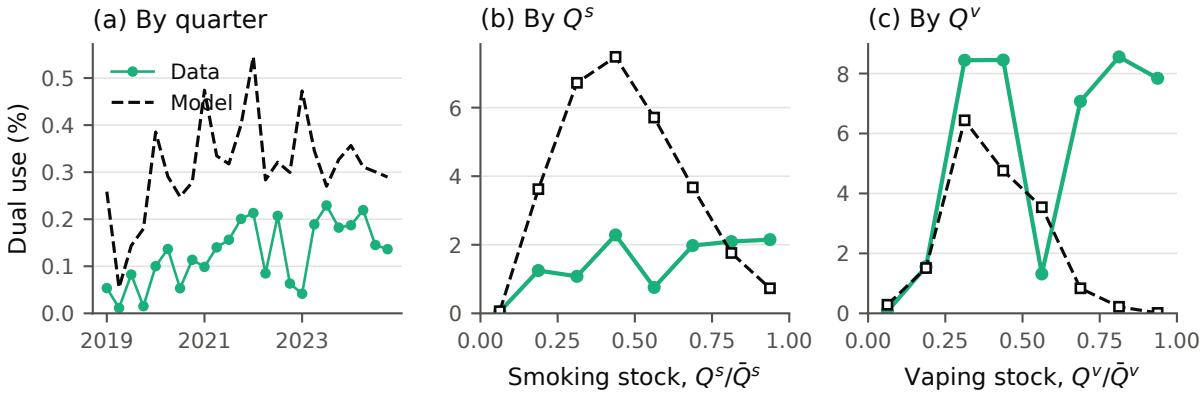


Notes: Share of household-months with a purchase of the product, observed and predicted.

Figure 23 shows the fit for dual use. The model over-predicts dual use: 0.32% of household-months against 0.13% in the data, and it does so in every year (0.16–0.38% against 0.04–0.17%). The excess comes from households with moderate cigarette stocks. At Q^s between 1.4 and 4.2, the model predicts dual use in 3.6–7.5% of household-months,

against 0.8–2.3% in the data. Among households with high vaping stocks the model under-predicts it instead: above $Q^v \approx 2.4$, 7–9% of household-months are dual use in the data and less than 1% in the model. The high-stock cells rest on few household-months (70 to 520 above $Q^v \approx 2.4$).

Figure 23: Model fit: dual use.

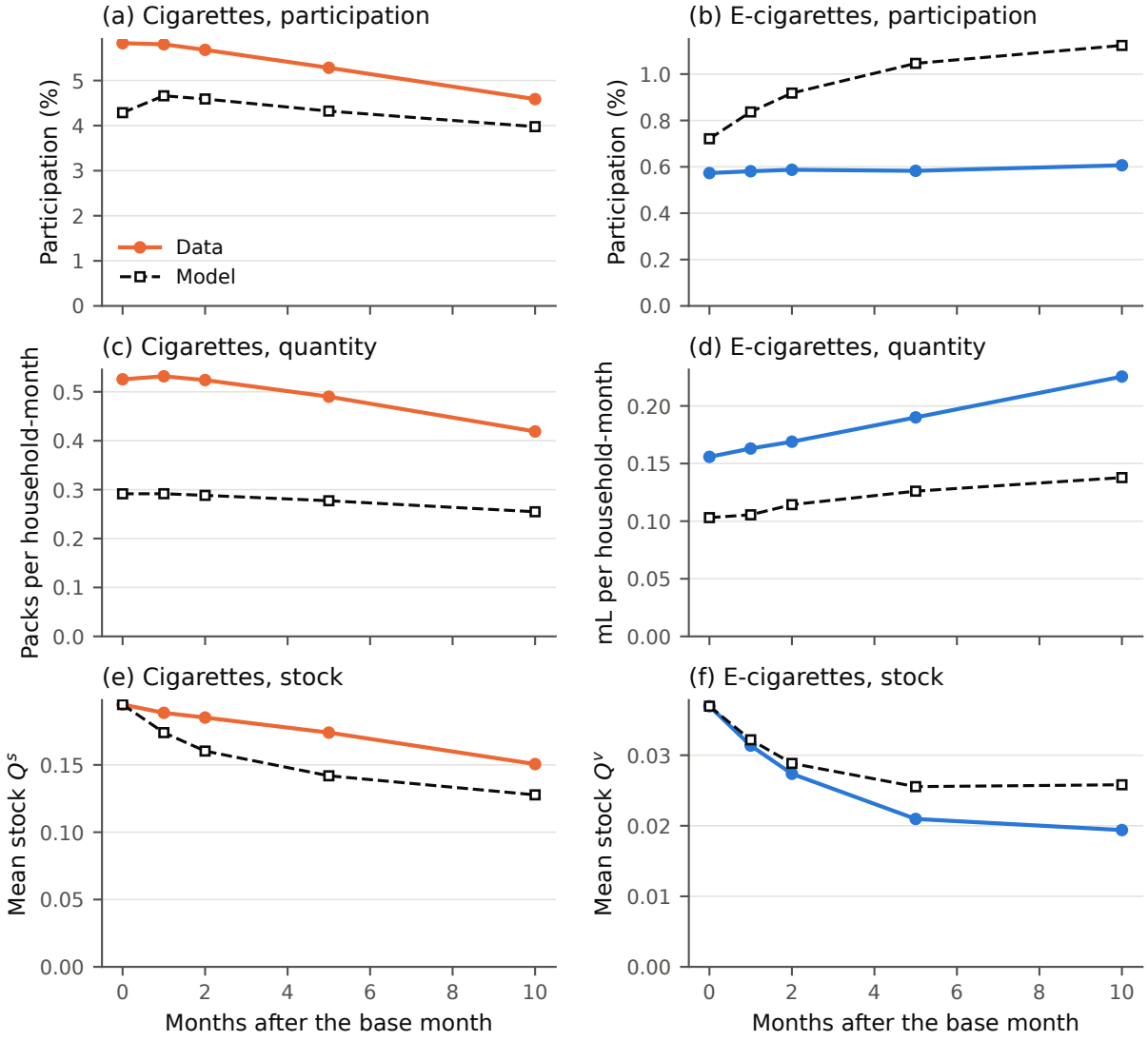


Notes: Share of household-months with purchases of both products, observed and predicted, (a) by quarter, (b) by the pre-action cigarette stock and (c) by the pre-action vaping stock.

Figure 24 assesses the dynamic fit. Starting from each of 62 base months (January 2019 to February 2024), the model is simulated forward for ten months along realised prices, and the simulated shares, quantities and stocks are compared with the same households in the data. At each horizon the comparison is restricted to households observed in every month from the base month on. Results are averaged over base months with equal weights.

On the participation margin the simulation carries the static errors forward (panels (a) and (b)). Smoking falls from 5.8% to 4.6% over ten months in this panel of continuously observed households, while the simulation starts at 4.3% and ends at 4.0%. Simulated vaping rises from 0.72% to 1.12%, while observed vaping stays near 0.6%. On the quantity margin the simulation is far off (panels (c)–(f)). Simulated purchases per household start at 0.29 packs and 0.10 mL a month, against 0.53 and 0.16 in the data. After ten months the mean cigarette stock is 0.128 against 0.151, and the mean e-cigarette stock 0.026 against 0.019. The simulated cigarette stock falls behind the data because simulated purchases are too small, while the simulated e-cigarette stock runs ahead because too many simulated households vape.

Figure 24: Dynamic fit.



Notes: By months after the base month, averaged over 62 base months. Panels (a)–(b): participation. Panels (c)–(d): monthly purchase per household, including zeros. Panels (e)–(f): mean pre-action stock, simulated by the model and implied by the observed purchases.